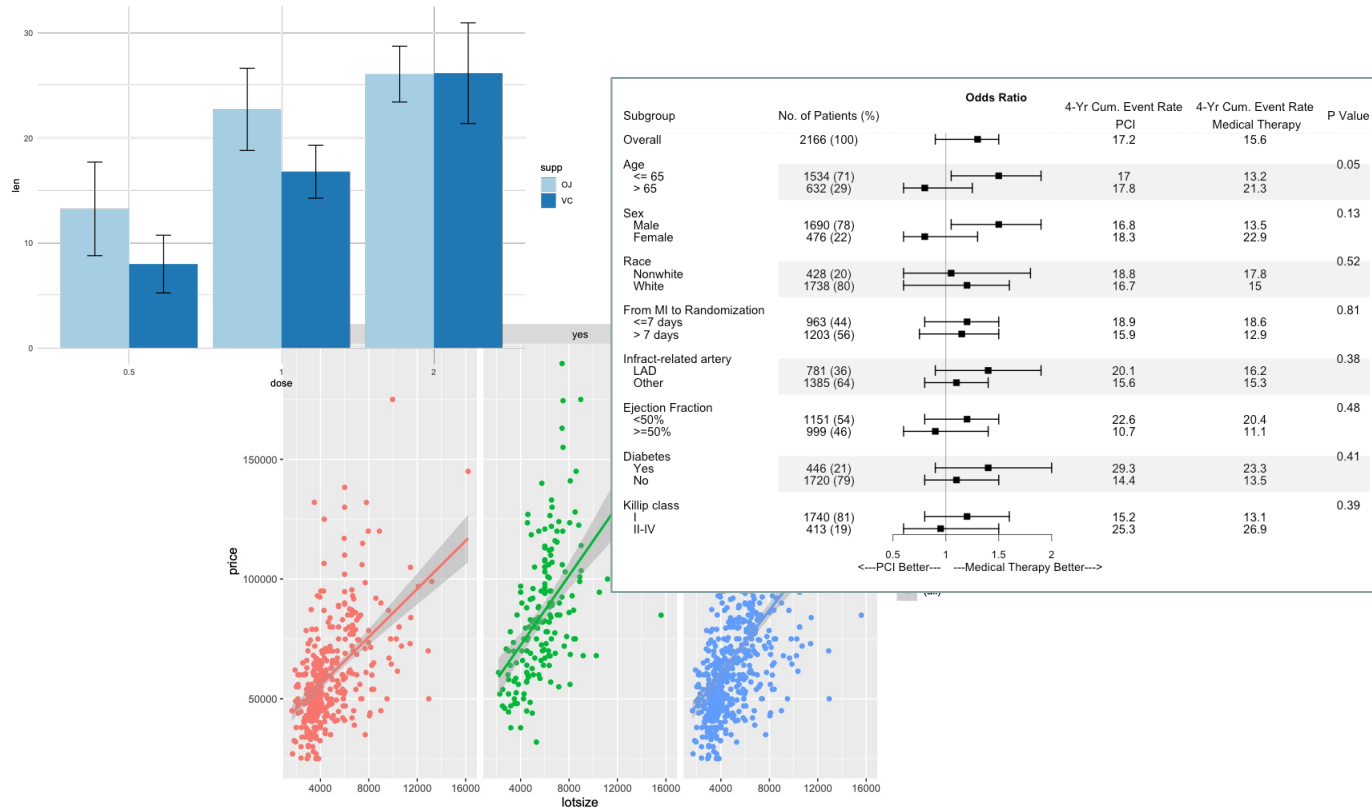


Introducción a



**“El año que novicio fui, espantome;
quíseme retirar, pero no hay cosa
que el tiempo y la costumbre no la dome”**

Arturo Pérez Reverte. *Corsarios de Levante*



Introducción

1. ¿Qué es R?
2. Características
3. ¿Por qué R?
4. Enlaces
5. Libros y material general
6. Screenshots
7. Comunidad R
8. Actualidad
9. Obtención e instalación
10. Documentación
11. Inicio de una sesión en R
12. Ayuda
13. R desde un editor

1. ¿Qué es R?

- R is a free software environment for statistical computing and graphics
- R is a language and environment for statistical computing and graphics.
- The R environment: R is an integrated suite of software facilities for data manipulation, calculation and graphical display.

R es un **lenguaje de programación** y un **entorno de software libre** diseñado para la computación estadística y el análisis de datos, que proporciona un conjunto integrado de herramientas para la manipulación y gestión de datos, la realización de cálculos estadísticos y la generación de representaciones gráficas.

Entorno

- el intérprete del lenguaje R
- funciones y paquetes para análisis estadístico
- herramientas de manipulación de datos
- capacidades de visualización gráfica
- mecanismos para extender el software mediante paquetes

1. ¿Qué es R?

- **R** es un lenguaje y entorno de programación **gratuito**
- Es un **lenguaje de programación orientado a objetos**.
- **Proyecto de software libre**, resultado de la implementación GNU del premiado lenguaje “S” y “Scheme”.
- R y S-Plus -versión comercial de S- son, probablemente, los dos lenguajes más utilizados en investigación por la comunidad estadística.
- Muy populares en el campo de la investigación biomédica, la bioinformática y las matemáticas financieras.

2. Características

- **Versatilidad:** posibilidad de cargar diferentes paquetes con finalidades específicas de cálculo o gráfico.
- El proyecto más conocido desarrollado sobre R es Bioconductor.
- Rmetrics está orientado al análisis de los mercados financieros y la valoración de instrumentos de inversión.
- R se distribuye bajo la licencia GNU GPL y está disponible para los sistemas operativos Windows, Macintosh, Unix y GNU/Linux.
- Algunas de las características más relevantes de R son que se trata de un lenguaje interpretado y que está orientado a objetos.

Versatilidad

Bayesian

Bayesian Inference

Cluster

Cluster Analysis & Finite Mixture Models

Econometrics

Computational Econometrics

Environmetrics

Analysis of Ecological and Environmental Data

ExperimentalDesign

Design of Experiments (DoE) & Analysis of Experimental Data

Finance

Empirical Finance

Genetics

Statistical Genetics

Graphics

Graphic Displays & Dynamic Graphics & Graphic Devices & Visualization

gR

gRaphical Models in R

MachineLearning

Machine Learning & Statistical Learning

Multivariate

Multivariate Statistics

NaturalLanguageProcessing

Natural Language Processing

Optimization

Optimization and Mathematical Programming

Pharmacokinetics

Analysis of Pharmacokinetic Data

Psychometrics

Psychometric Models and Methods

Robust

Robust Statistical Methods

SocialSciences

Statistics for the Social Sciences

Spatial

Analysis of Spatial Data

Survival

Survival Analysis

pbdR: **programming with big data in R**

Simplifying Scalability

[About](#)[Packages](#)[Download](#)[Install](#)[Tutorials](#)[FAQ](#)[Misc](#)

Overview

The "Programming with Big Data in R" project (pbdR) enables high-level distributed data parallelism in R, so that it can easily utilize large HPC platforms with thousands of cores, making the R language scale to unparalleled heights. We interpret big data quite literally to mean that its size requires parallel processing either because it does not fit in the memory of a single multicore machine or because we need to make its processing time tolerable.

We achieve this, in part, by providing a simple interface to scalable, high performance libraries, such as [MPI](#), [ScaLAPACK](#), and [NetCDF4](#). The routines in these libraries are engaged through R's classes and methods, so that the R language syntax is largely preserved, but with new, scalable, compiled code underneath. Most of the cumbersome distributed details are abstracted away for the user, although they are readily accessible should the user desire them.

We provide [several R packages](#) which focus on analyzing big data, especially on large distributed machines. The packages are meant to be used in the [Single Program/Multiple Data \(SPMD\)](#) programming model ([see also](#)), with special focus on large scale computing clusters. Packages and instructions for installation are available on the [downloads page](#).

<http://www.r-pbd.org/>


RDataMining.com: R and Data Mining

Search this site

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Training

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Documents

Introduction to Data Mining with R

R Reference Card for Data Mining

R and Data Mining: Examples and Case Studies

Introduction to Data Mining with R and Data Import/Export in R

Data Exploration and Visualization with R

Regression and Classification with R

Data Clustering with R

Association Rule Mining with R

Text Mining with R: Twitter Data Analysis

Time Series Analysis and Mining with R

Examples

Data Exploration

Decision Trees

Random Forest

k-means Clustering

Hierarchical Clustering

Outlier Detection

Time Series Forecasting

Time Series Analysis

Time Series Clustering and Classification

Association Rules

Text Mining

Twitter Follower Map

Social Network Analysis

Multidimensional Scaling (MDS)

Principal Component Analysis (PCA)

Parallel Computing

Other Examples

Big Data

Big Data Platforms

This website presents documents, examples, tutorials and resources on R and data mining.

Documents on Data Mining with R

- [R Reference Card for Data Mining](#)
- [R and Data Mining: Examples and Case Studies](#)
- [Introduction to Data Mining with R](#)
- [RDataMining slides series on](#)
 - [Introduction to Data Mining with R and Data Import/Export in R](#)
 - [Data Exploration and Visualization with R,](#)
 - [Regression and Classification with R,](#)
 - [Data Clustering with R,](#)
 - [Association Rule Mining with R,](#)
 - [Text Mining with R: Twitter Data Analysis,](#) and
 - [Time Series Analysis and Mining with R](#)

Examples on Data Mining with R

- [data exploration](#)
- [decision trees](#)
- [k-means clustering](#) and [hierarchical clustering](#)
- [outlier detection](#)
- [time series decomposition and forecasting](#)
- [time series clustering and classification](#)
- [association rules](#)
- [text mining](#)
- [social network analysis](#)
- [Multidimensional Scaling \(MDS\)](#)
- [parallel computing](#)
- [many examples from other websites](#)







News

[www2.rdatamining.com: a mirror site for Chinese users](#)

RDataMining.com now has a mirror website at <http://www2.rdatamining.com>. Users in China can download RDataMining documents, code and data at above mirror site, if no access to ...

Posted Aug 5, 2015, 5:31 AM by Yanchang Zhao

Showing posts 1 - 1 of 73. [View more »](#)

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 Sydney, 8-9 August 2015





**Bioconductor**
OPEN SOURCE SOFTWARE FOR BIOINFORMATICS

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Home » [Help](#) » [Publications](#) » Bioinformatics and Computational Biology Solutions Using R and Bioconductor



Bioinformatics and Computational Biology Solutions Using R and Bioconductor

Editors: [Robert Gentleman](#), [Vince Carey](#), [Wolfgang Huber](#), [Rafael Irizarry](#), [Sandrine Dudoit](#)

Released 1 Sept 2005

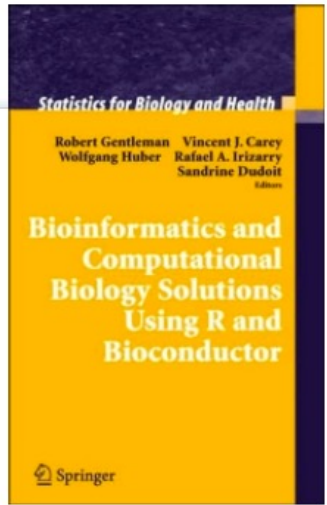
Contents:

- [Preprocessing data from genomic experiments](#) (eds W. Huber, R. A. Irizarry)
- [Metadata resources: annotating and visualizing genomic data and analyses](#) (V. J. Carey, R. Gentleman eds)
- [Analysis of microarray experiments](#) (eds S. Dudoit, V. J. Carey)
- [Graphs and networks for genomic data analysis](#) (R. Gentleman, V. J. Carey eds)
- [Case studies](#) (eds. S. Dudoit, R. Irizarry)

[Getting the Monograph via R](#)

[Data](#)

To purchase, online orders can be made through [Springer Online](#) or through [Barnes and Noble Online](#).



<http://www.bioconductor.org/help/publications/books/bioinformatics-and-computational-biology-solutions/>

2. Características

- Puede integrarse con distintas bases de datos
- Existen paquetes que facilitan su utilización desde lenguajes de programación interpretados como Perl y Python o en lenguajes de código compilado, por ejemplo, C o Fortran.
- Una amplia colección de paquetes se encuentran en CRAN.
- Existen diversas interfaces gráficas para R, como R-Commander.
- Se parece a Matlab y a Octave, y su sintaxis recuerda a C/C++.

3. ¿Por qué R?

- **No hay razón en cambiar si se realizan tareas sencillas ya implementadas y no se intentará nada nuevo en el futuro**
- **Principal razón para cambiar a R:** *cobertura de análisis y disponibilidad de nuevas aplicaciones* en campos por ejemplo de modelos lineales generalizados.
- Ser capaz de entender la literatura.
- La mayoría de “top people” en las distintas disciplinas lo usa.

3. ¿Por qué R?

- Una gran parte de los estadísticos lo utiliza.
- Gran soporte y seguro.
- Una gran red de respaldo.
- Capacidad de programar y construir funciones propias
- Es **GRATIS**

4. Enlaces

- El proyecto R:

<http://www.r-project.org/>

- The R Reference Manual - Base Package by the R Development Core Team. ISBN 0-9546120-0-0 (vol. 1), ISBN 0-9546120-1-9 (vol. 2) :

<http://www.network-theory.co.uk/R/base/>

- Colección librerías en CRAN (Comprehensive R Archive Network):

<https://cran.r-project.org/>



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[Bioconductor](#)

[Related Projects](#)

The R Project for Statistical Computing

Getting Started

R is a free software environment for statistical computing and graphics. It compiles and runs on a wide variety of UNIX platforms, Windows and MacOS. To [download R](#), please choose your preferred CRAN mirror.

If you have questions about R like how to download and install the software, or what the license terms are, please read our [answers to frequently asked questions](#) before you send an email.

News

- **R version 3.2.2 (Fire Safety) prerelease versions** will appear starting 2015-08-04. Final release is scheduled for 2015-08-14.
- **The R Journal Volume 7/1** is available.
- **R version 3.2.1 (World-Famous Astronaut)** has been released on 2015-06-18.
- **R version 3.1.3 (Smooth Sidewalk)** has been released on 2015-03-09.
- **useR! 2015**, will take place at the University of Aalborg, Denmark, June 30 - July 3, 2015.
- **useR! 2014**, took place at the University of California, Los Angeles, USA June 30 - July 3, 2014.

CRAN Mirrors

The Comprehensive R Archive Network is available at the following URLs, please choose a location close to you. Some statistics on the status of the mirrors can be found here: [main page](#), [windows old release](#), [windows old release](#).

0-Cloud	https://cran.rstudio.com/ http://cran.rstudio.com/	Rstudio, automatic redirection to servers worldwide Rstudio, automatic redirection to servers worldwide
Algeria	http://cran.usbhb.dz/	University of Science and Technology Houari Boumediene
Argentina	http://mirror.fcaglp.unlp.edu.ar/CRAN/	Universidad Nacional de La Plata
Australia	http://cran.csiro.au/ http://cran.ms.unimelb.edu.au/	CSIRO University of Melbourne
Austria	https://cran.r-project.org/ http://cran.at.r-project.org/	Wirtschaftsuniversität Wien Wirtschaftsuniversität Wien
Belgium	http://www.freestats.org/cran/	K.U.Leuven Association
Brazil	http://hbcgib.usesc.br/mirrors/cran/ http://cran-fc3sl.ufpe.br/ http://cran.fiocruz.br/ http://www.vps.fmvz.usp.br/CRAN/ http://brieger.esbq.usp.br/CRAN/	Center for Comp. Biol. at Universidade Estadual de Santa Cruz Universidade Federal do Paraná Oswaldo Cruz Foundation, Rio de Janeiro University of Sao Paulo, Sao Paulo University of Sao Paulo, Piracicaba
Canada	http://cran.stat.sfu.ca/ http://mirror.its.dal.ca/cran/ http://cran.utoronto.ca/ http://cran.skaakaforyou.com/ http://cran.parentingamerica.com/	Simon Fraser University, Burnaby Dalhousie University, Halifax University of Toronto iWeb, Montreal iWeb, Montreal
Chile	http://dichilestat.puc.cl/	Pontificia Universidad Catolica de Chile, Santiago
China	http://mirror.bjtu.edu.cn/cran/ http://mirrors.opencas.cn/cran/ http://mirrors.tuna.tsinghua.edu.cn/CRAN/ http://mirrors.tuna.tsinghua.edu.cn/CRAN/ http://mirrors.usc.edu.cn/CRAN/ http://mirrors.usc.edu.cn/CRAN/ http://mirrors.xmu.edu.cn/CRAN/	Beijing Jiaotong University, Beijing Chinese Academy of Sciences, Beijing TUNA Team, Tsinghua University TUNA Team, Tsinghua University University of Science and Technology of China University of Science and Technology of China Xiamen University
Colombia	http://www.icsi.edu.co/CRAN/ http://www.icsi.edu.co/CRAN/	Icesi University Icesi University



CRAN

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The Comprehensive R Archive Network

Download and Install R

Precompiled binary distributions of the base system and contributed packages, **Windows** and **Mac** users most likely want one of these versions of R:

- [Download R for Linux](#)
- [Download R for \(Mac\) OS X](#)
- [Download R for Windows](#)

R is part of many Linux distributions, you should check with your Linux package management system in addition to the link above.

Source Code for all Platforms

Windows and Mac users most likely want to download the precompiled binaries listed in the upper box, not the source code. The sources have to be compiled before you can use them. If you do not know what this means, you probably do not want to do it!

- The latest release (2015-06-18, World-Famous Astronaut) [R-3.2.1.tar.gz](#), read [what's new](#) in the latest version.
- Sources of [R alpha](#) and [beta releases](#) (daily snapshots, created only in time periods before a planned release).
- Daily snapshots of current patched and development versions are [available here](#). Please read about [new features and bug fixes](#) before filing corresponding feature requests or bug reports.
- Source code of older versions of R is [available here](#).
- Contributed extension [packages](#)

Questions About R

- If you have questions about R like how to download and install the software, or what the license terms are, please read our [answers to frequently asked questions](#) before you send an email.

What are R and CRAN?

R is "GNU S", a freely available language and environment for statistical computing and graphics which provides a wide variety of statistical and graphical techniques: linear and nonlinear modelling, statistical tests, time series analysis, classification, clustering, etc. Please consult the [R project homepage](#) for further information.

CRAN is a network of ftp and web servers around the world that store identical, up-to-date, versions of code and documentation for R. Please use the CRAN [mirror](#) nearest to you to minimize network load.

Submitting to CRAN

To "submit" a package to CRAN, check that your submission meets the [CRAN Repository Policy](#) and then use the [web form](#).

If this fails, upload to [ftp://CRAN.R-project.org/incoming/](#) and send an email to CRAN@R-project.org following the policy. Please do not attach submissions to emails, because this will clutter up the mailboxes of half a dozen people.

Note that we generally do not accept submissions of precompiled binaries due to security reasons. All binary distribution listed above are compiled by selected maintainers, who are in charge for all binaries of their platform, respectively.

This server is hosted by the [Institute for Statistics and Mathematics of WU \(Wirtschaftsuniversität Wien\)](#).

4. Enlaces

- R-Wiki :

<http://wiki.r-project.org/rwiki/doku.php>

- Interfaz Web para R :

<http://www.math.montana.edu/Rweb/>

- R Graph Gallery :

<http://addictedtor.free.fr/graphiques/>

RELATED SITES: [R-project](#) | [CRAN](#) | [Bioconductor](#) | [R Movies Gallery](#) | [R Wiki](#) | [SODV](#) | [Graphics CTV](#)

search :

[Home](#) | [Browse](#) | [Related](#) | [Source code](#) | [Graphics List](#) | [Thumbnails](#)

Welcome!

The R Graph Gallery aims to present several different graphics fully created with the programming environment [R](http://www.r-project.org/) (<http://www.r-project.org/>). Graphs are gathered in a MySQL database and browsable thanks to PHP.

We hope that this gallery will provide many benefits, including:

- Discover new graphics that are suited to specific situations
- Highlight the power/Rui graphical abilities of R
- Share coding experiment

How to support the gallery

... here are a few ideas

- Make a link from your website/blog.
- Send a graph (source code + comments), or suggestions to improve the website: by [e-mail](#) or [wiki](#)

Enter the gallery

- 3 best ranked graphs
- 3 last included graphs
- 3 random selected graphs
- 3 last seen graphs

R Project

R is a system for statistical computation and graphics.

It consists of a language plus a run-time environment with graphics, a debugger, access to certain system functions, and the ability to run programs stored in script files.

R Graphic Engine

One of R's strengths is the ease with which well-designed publication-quality plots can be produced, including mathematical symbols and formulae where needed. Great care has been taken over the defaults for the minor design choices in graphics, but the user retains full control.

Thanks

Thanks to the R Core Team for the wonderful work made on our favorite tool.

Technology

This site is coded in xHTML with CSS. It uses MySQL and PHP. [The Simg](#) has been used for the design.

Related Websites

- [R project](#)
- [CRAN](#)
- [Bioconductor](#)
- [r-spatial](#)
- [R Techniques and Graphics Gallery](#)
- [Paul Murrell's book](#)
- [Michael Friendly's Gallery of Data Visualization : The Best and Worst of Statistical Graphics.](#)



Statistical Analysis On The Web

Rweb is a Web based interface to **R** (a statistical analysis package) that takes the submitted code, runs **R** on the code (in batch mode), and returns the output (printed and graphical). If you have any comments, criticisms, bug reports, or if you're just lonely you can email me ([Jeff Banfield](mailto:Jeff.Banfield@math.montana.edu)) at [Jeff Banfield](mailto:Jeff.Banfield@math.montana.edu) at [Jeff Banfield](mailto:Jeff.Banfield@math.montana.edu). If you are interested in setting up your own Rweb server visit the [Rweb Resources Page](#) to download the source code and find links to supporting software.

R is a free statistical analysis package developed by Robert Gentleman and Ross Ihaka. It is almost source code compatible with S or Splus so if you have experience with those languages you should have no problem using **Rweb**. For information about **R** you can read the [R introduction page](#), contact the [R authors](#), or visit the [R Archive](#). If you would like to learn more about using **R** you can try a fairly rough latex2html conversion of the [R notes](#) developed by Venables and Smith and then modified for **R** by Gentleman and Ihaka.

Which Version Do You Want?

Rweb

A simple web interface to **R** that works on most browsers and requires knowledge of **R** (or Splus). You type in the code, click the submit button and a page with the results (analysis and graphs) is returned. This version of **Rweb** only requires a browser that can handle forms and graphics (if you do any plotting). There is a small code snippet at the end of the page in case you don't know **R** but you still want to try it out.

JavaScript Version of Rweb

This is the same **Rweb** as above but with a few display enhancements based on JavaScript. If your browser understands JavaScript this version of **Rweb** should work for you.

Rweb modules

These modules are designed as a point and click forms based interface to **R** for use in introductory statistics courses. Students do not need to know anything about writing programs in **R**. They choose a data set, the type of analysis, and the options for the analysis. The currently implemented analyses include Regression, Summary statistics and graphs, ANOVA, Two Way Tables, and a Probability calculator. The output includes printouts and graphs which can be cut and pasted into project writeups or printed out. I'm making the modules up as I go (both content and design) and I would love to have some advice. So, anybody that has an idea about improving these modules, please let me know.

© 1998 [Jeff Banfield](#)
 Author: [Jeff Banfield](#)
 Email: Jeff.Banfield@math.montana.edu

Last Modified: 25-Jun-1999

[[start]]

data(trees)
trees.lm <- lm(volume ~ girth + height, data=trees, weights=1/area)
summary(trees.lm)

Recent changes

Search

Trace: > start

R Wiki - Overview

R is a free software environment for statistical computing and graphics. It runs on a wide variety of UNIX platforms, Windows and MacOS. This R Wiki is dedicated to the collaborative writing of R documentation.

For information on browsers, RSS syndication, copyright, ... [read usage](#).

R comes with several **official** manuals and **FAQs**. These should be your **primary source of information**. Note also that the main location for general questions about **R** is the **R-Help** mailing list (but please make sure to read the **posting guide** before posing a question there!). Finally, **R-News** is a newsletter dedicated to **R**.

Table of contents

(click on a item to expand)

- Getting started
- Guides
- Student Section
- Tips & Tricks
- Links
- R graphs, packages and documentation
- Users, Wiki, etc.

start not - Last modified: 2008/04/03 by [Kevin](#)

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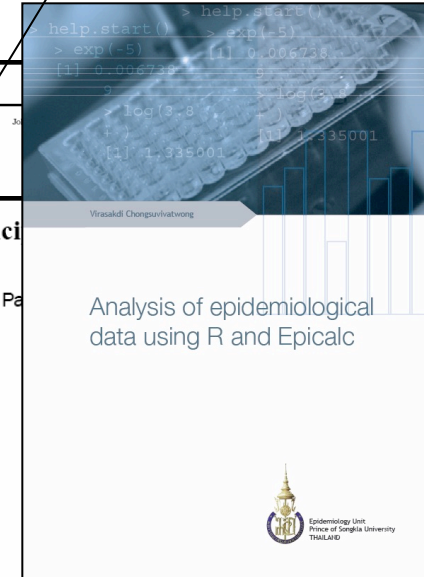
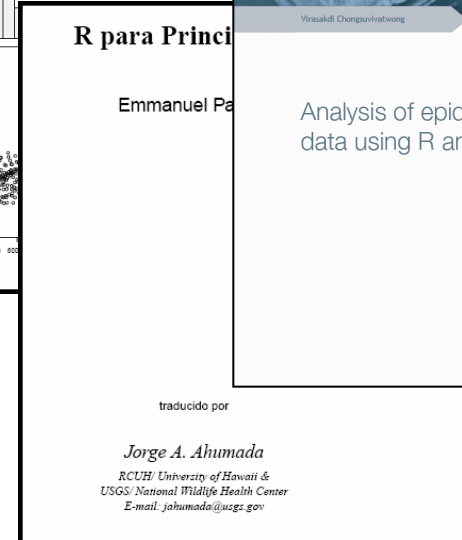
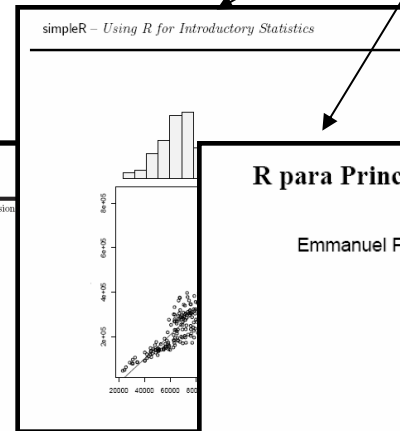
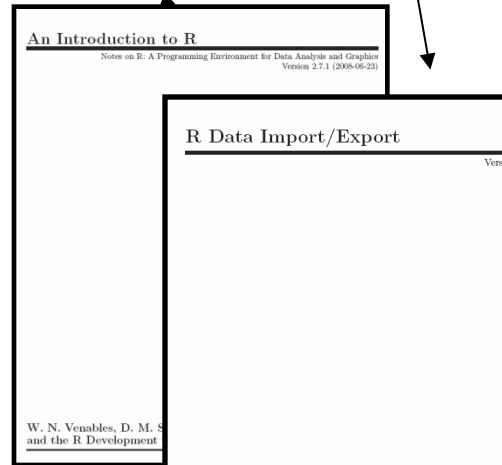
R Wiki powered by [R-CHTML](#) and optimized for Firefox

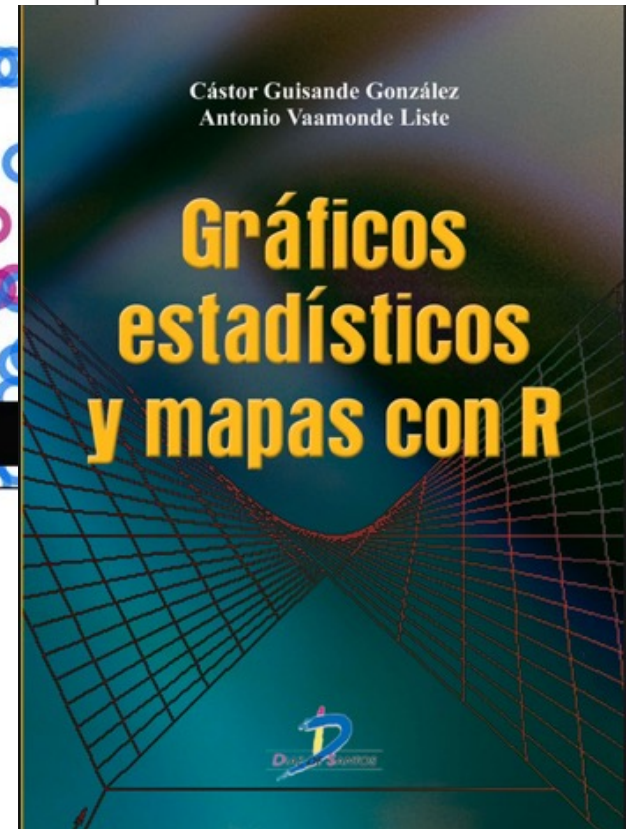
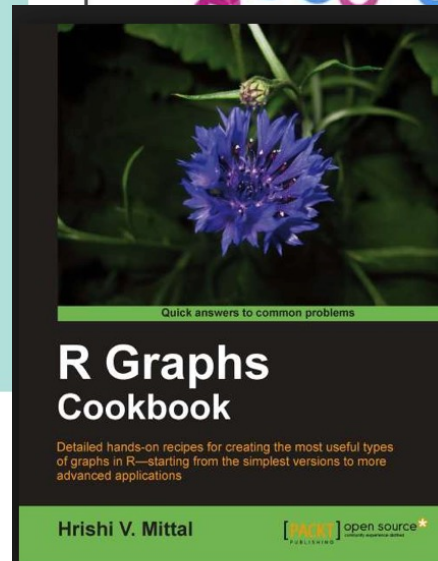
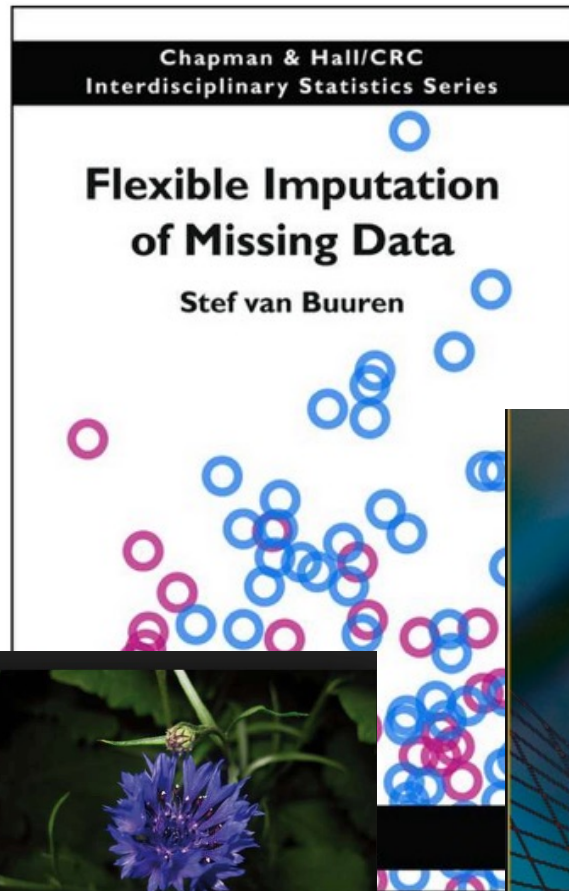
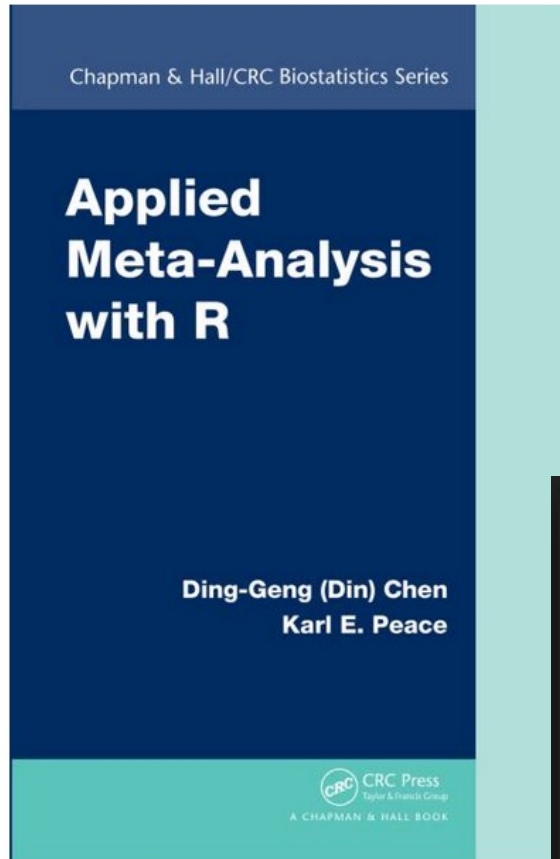
5. Libros y material general

Libros

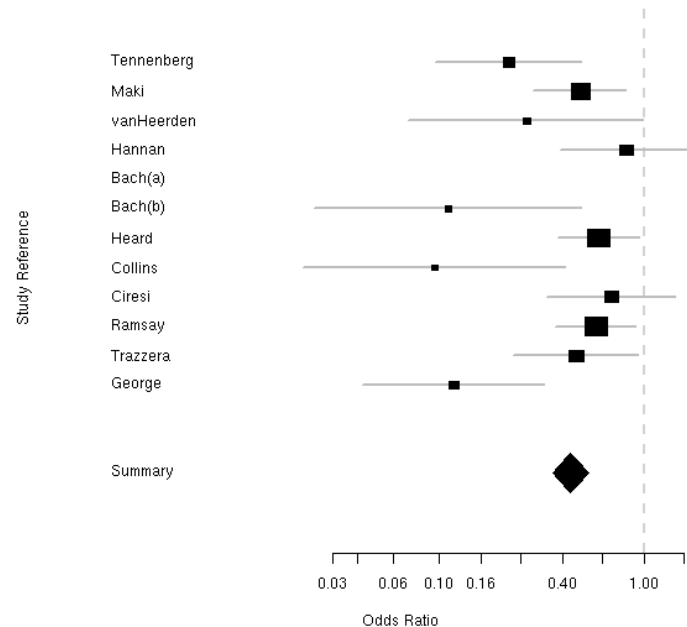
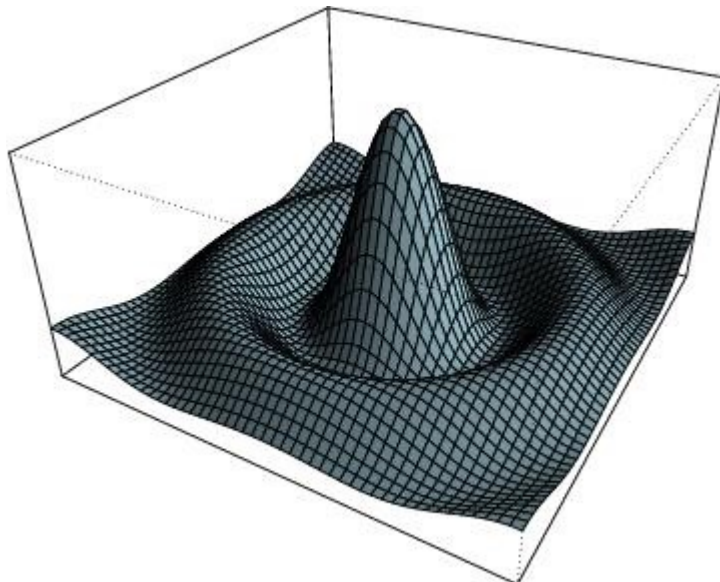
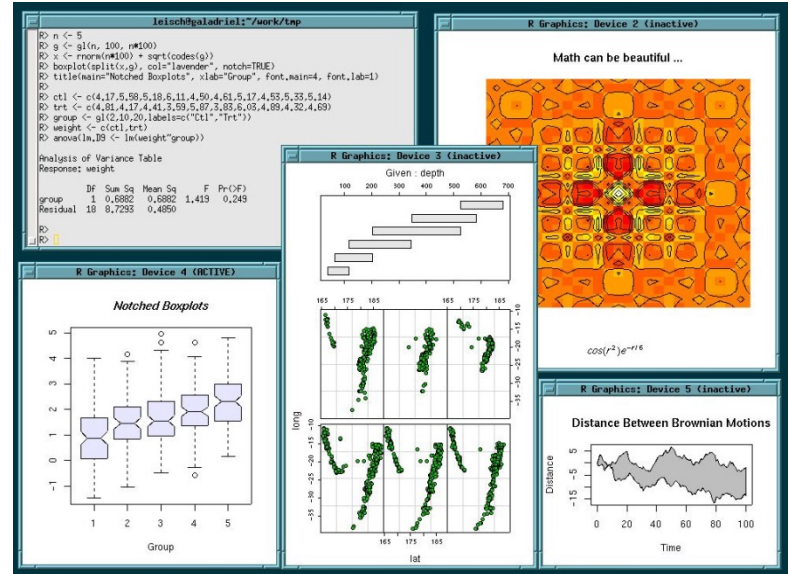
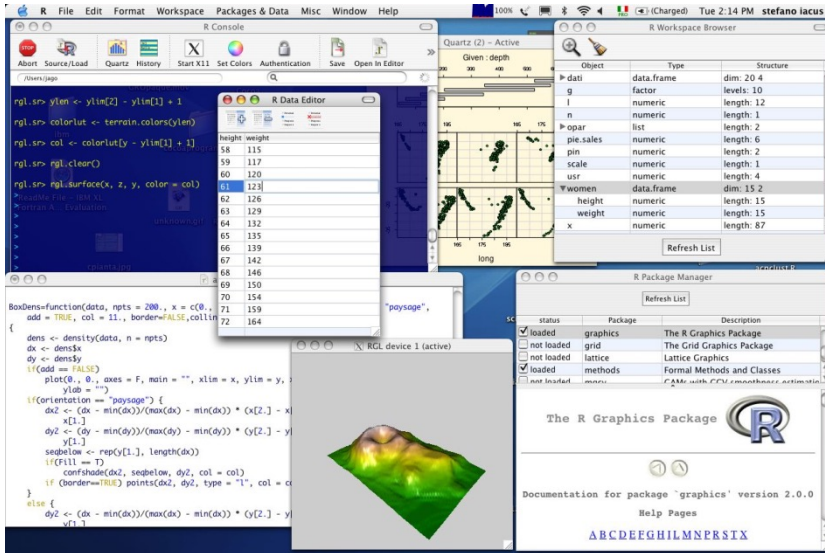
Manuales *R Core Team*

Documentación





6. Screenshots



7. Comunidad R

- Burns Statistics Ltd., London, U.K.
- Department of Statistics, Brigham Young University, Utah, USA
- Paul von Eikeren, USA
- Institute of Mathematical Statistics (IMS), Ohio, USA
- Loyalty Matrix Inc., California, USA
- Mango Solutions, Chippenham, UK
- Marc Schwartz, USA
- Merck and Co. Inc., USA
- Numbers Internation Pty Ltd, Australia
- Prediction Company, Santa Fe, New Mexico, USA
- Saxo Bank, Denmark
- Schröder Investment Management Ltd., London, UK
- InterContinental Hotels Group, USA
- Shell Statistics and Chemometrics, Chester, UK
- Statisticon AB, Uppsala, Sweden
- Astra Zeneca R&D Mölndal, Mölndal, Sweden
- AT&T Labs, New Jersey, USA
- Baxter AG, Vienna, Austria
- Baxter Healthcare Corp., California, USA
- BC Cancer Agency, Vancouver, Canada
- Black Mesa Capital, Santa Fe, USA
- Boehringer Ingelheim Austria GmbH, Vienna, Austria
- Breast Center at Baylor College of Medicine, Houston, Texas, USA
- Center für digitale Systeme, Freie Universität Berlin, Germany
- Dana-Farber Cancer Institute, Boston, USA
- Department of Biostatistics, Johns Hopkins University, Maryland, USA
- Department of Biostatistics, Vanderbilt University School of Medicine, USA
- Department of Economics, Stockholm University, Sweden
- Department of Mathematics and Statistics, Utah State University, USA
- Department of Statistics, University of California at Los Angeles, USA
- Department of Statistics, University of Warwick, Coventry, UK
- Department of Statistics, University of Wisconsin-Madison, Wisconsin, USA
- Department of Statistics, Iowa State University, USA
- Department of Statistics & Actuarial Science, University of Iowa, USA
- Dipartimento di Statistica, Università Cà Foscari di Venezia, Italy
- Division of Biostatistics, University of California, Berkeley, USA
- Ef-prime Inc., Tokyo, Japan
- European Bioinformatics Inst., UK
- Hygeia Associates, California, USA
- Lehrstuhl für Rechnerorientierte Statistik und Datenanalyse, University of Augsburg, Germany
- MPI for Demographic Research, Rostock, Germany
- Massachusetts General Hospital Biostatistics Center, Boston, USA
- National Public Health Institute, Helsinki, Finland
- Norwegian Institute of Marine Research, Bergen, Norway
- School of Economics and Finance, Victoria University of Wellington, New Zealand
- Spotfire, Massachusetts, USA
- TERRA Lab, University of Regina - Department of Geography, Canada
- ViaLactia Biosciences (NZ) Ltd, Auckland, New Zealand
- Adelchi Azzalini (Italy)
- AT&T Research (USA)
- Austrian Association for Statistical Computing (Austria)
- BC Cancer Agency, Vancouver (Canada)
- Fabian Barth (Germany)
- Biostatistics and Research Decision Sciences, Merck Research Laboratories (USA)
- Brian Caffo (USA)
- David W. Crawford (USA)
- Dianne Cook (USA)
- Yves DeVillè (France)
- Department of Economics, University of Milano (Italy)
- Dipartimento di Scienze Statistiche e Matematiche di Palermo (Italy)
- Emanuele De Rinaldis (Italy)
- Zubin Dowlaty (USA)
- Faculty of Economics, University of Groningen (Netherlands)
- Jaimison Fargo (USA)

8. Actualidad

- El proyecto R mantiene dos series de Conferencias organizadas regularmente por los miembros de la Comunidad de R:
 - 1) **useR!**== forum de la comunidad de usuarios de R.
 - 2) **DSC**== una plataforma de desarrollo de software estadístico.





International R User Conference

- Usuarios y desarrolladores de R.
- Presentaciones de invitados y usuarios contribuidores.
- Abstracts de usuarios

- Serie histórica:

useR! 2004, Vienna, Austria
useR! 2006, Vienna, Austria
useR! 2007, Ames, IA, USA
useR! 2008, Dortmund, Germany
useR! 2009, Rennes, France
useR! 2010, Gaithersburg, MD, USA
useR! 2011, Coventry, UK
useR! 2012, Nashville, TN, USA
useR! 2013, Albacete, Spain
useR! 2014, Los Angeles, CA, USA
useR! 2015, Aalborg, Denmark.
useR! 2016, Stanford, CA, USA
useR! 2017, Brussels, Belgium.
useR! 2018, Brisbane, Australia.
useR! 2019, Toulouse, France.
useR! 2020, online

useR! 2025, Durham, NC, USA.
useR! 2021, online.
useR! 2022, online.
useR! 2024, Salzburg, Austria.
useR! 2025, Durham, NC, USA.
useR! 2026. Warsaw, Poland.



The R User Conference 2008
 August 12-14, Technische Universität Dortmund,
 Germany

Organizer: Fakultät Statistik, Technische Universität Dortmund
 Co-Organizer: Austrian Association for Statistical Computing
 Sponsors: R Foundation for Statistical Computing



A STAR ALLIANCE MEMBER



Conference

About the Conference
 Date & Location
 Important Dates
 Call for Papers
 Download: Logo, Flyer, Poster
 Funding
 Participants
 Photos

Program

Conference Program
 Download (PDF): Program and Book of abstracts
 Invited Lectures
 Presentations (including slides!)
 Tutorials
 Social Program
 Program Committee

Dortmund

Accommodation
 About Dortmund
 Travel information

About the Conference

useR! 2008, the R user conference, takes place at the Fakultät Statistik, Technische Universität Dortmund, Germany from 2008-08-12 to 2008-08-14. Pre-conference tutorials take place on August 11. The conference is organized by the Fakultät Statistik, Technische Universität Dortmund and the Austrian Association for Statistical Computing (AASC). It is funded by the R Foundation for Statistical Computing.

Following the successful **useR! 2004**, **useR! 2006**, and **useR! 2007** conferences, the conference is focused on

1. R as the 'lingua franca' of data analysis and statistical computing,
2. providing a platform for R users to discuss and exchange ideas how R can be used to do statistical computations, data analysis, visualization and exciting applications in various fields,
3. giving an overview of the new features of the rapidly evolving R project.

As for the predecessor conference, the program consists of two parts:

1. invited lectures discussing new R developments and exciting applications of R,
2. user-contributed presentations reflecting the wide range of fields in which R is used to analyze data.

A major goal of the useR! conference is to bring users from various fields together and provide a platform for discussion and exchange of ideas: both in the formal framework of presentations as well as in the informal part of the conference in Dortmund's famous beer pubs and restaurants.

Prior to the conference, on 2008-08-11, there are **tutorials** offered at the conference site. Each tutorial has a length of 3 hours and takes place either in the morning or afternoon.

Organizing Team:

Uwe Ligges (conference), Achim Zeileis (program), Claus Weihs, Gerd Kopp (local organization), Friedrich Leisch, Torsten Hothorn

Program committee:

Date & Location

August 12-14, 2008 (iCalendar file)

Fakultät Statistik
 Technische Universität Dortmund
 Vogelpothsweg 87
 44227 Dortmund
 Germany

Important Dates

2008-03-31
 early registration deadline
 2008-03-31
 submission deadline of abstracts
 2008-05-18
 notification of acceptance
 2008-05-31 (CEST)
 regular registration deadline
 closed now:
 online registration
 2008-07-25
 registration deadline
 (later registration possible on site)
 2008-08-11
 tutorials
 2008-08-12
 conference start
 2008-08-14
 conference end

Click [A] for abstracts or [S] for slides.

Author	Title
[A] Thomas Achia, Atinuke Adebaniji, John Owino, Anne Wangombe	Spatial Durbin Model for Poverty Mapping and Analysis
[A][S] Daniel Adler, Jens Oehlschlägel, Oleg Nenadic, Walter Zucchini	Large atomic data in R: package 'ff'
[A] Claudio Agostinelli	Robust Inference in Generalized Linear Models
[A] Arthur Allignol, Jan Beyersmann, Martin Schumacher	mvna, a R-package for the Multivariate Nelson-Aalen Estimator in Multistate Models
[A] Micah Altman, Michael McDonald	BARD: Better Automated Redistricting
[A] Gianmarco Altoè	The 'deltaR' package: a flexible way to compare regression models on independent samples using a bootstrap approach
[A] M. Rui Alves, M. Beatriz Oliveira	Automatic construction of graphical outputs of common multivariate analyses with a special reference to predictive biplots
[A][S] Thomas Baier	R in Automation: Accessing Real-time-data
[A] Mehmet Balciyar	RSTAR: A Package for Smooth Transition Autoregressive Modeling Using R
[A][S] Marco Ballin, Giulio Barcaroli	Tree-based and GA tools for optimal sampling design
[A][S] Susana Barbosa	ArDec: Autoregressive-based time series decomposition in R
[A][S] Benjamin Barnes, Karen Steindorf	Visualizing multivariate categorical and continuous data from epidemiologic studies: An expanded scatter plot matrix
[A][S] Jan Beyersmann, Arthur Allignol, Martin Schumacher	Understanding product integration
[A][S] Thomas Binsl, Jaap Heringa, David Alders, Hans van Beek	FluxES: A Network

ESTIMATION OF STANDARD ERRORS IN NON-LINEAR REGRESSION MODELS: SPATIAL VARIATION IN RISK AROUND PUTATIVE SOURCES

Ramis, Rebeca^{1,2,3}; Diggle, Peter¹; López-Abente, Gonzalo^{2,3}

¹ Department of Medicine, Lancaster University, UK.

² Cancer and Environmental Epidemiology Area, National Centre for Epidemiology. Carlos III Institute of Health, Madrid, Spain.

³ CIBERESP

Background

We consider the problem of investigating spatial variation in the risk of non-infectious diseases in populations exposed to pollution from one or more point sources.

The data most commonly available to study this question include case-counts (O_i) in each of a set of areas that partition the geographical region of interest, suitable denominators, E_i , proportional to the expected number of cases in each area, and the locations of the relevant point sources, from which we can compute distances d_{ij} between the j th focus and a reference location, typically the centroid, within the i th area. Also available in most applications are covariates relating to socio-economic status or other risk-factors associated with each area, which we denote by Z_k .

The standard approach to the analysis of data of this kind is a log-linear regression of the case-counts on the covariates, with log-transformed denominators as an offset variable. To model distance-related point source effects, a log-linear formulation is unrealistic because of the need to combine an elevated risk close to the source with a neutral long-distance effect. We therefore extend the model by including a non-linear distance function, $f(d_{ij})$,

Segmented Poisson Models

Vidal E^{1,2}, Pastor-Barriuso R^{1,2}, Pollan M^{1,2}, Lopez-Abente G^{1,2}.

¹ Environmental and Cancer Epidemiology Unit, National Centre for Epidemiology and Public Health, Madrid, Spain.

² CIBERESP, Spain.

Standard dose-response analyses (such as categorical, spline, or nonparametric) provide flexible tools to describe the overall shape of the dose-response relation across the range of exposure, but the identification of trend changes with these methods is subjective. Specifically, we propose to formally test for the existence of change-points in risk trends.

We propose a log-linear model for aggregated data with Poisson variance and in which the predictor function consists of two intersecting straight lines connected at a change-point through a hyperbolic transition function, that allows for abrupt changes of slope between the linear trends. The model, that was implemented as an R function, provides estimates of the existence of a change-point, as well as point and interval estimates for its location and above it.



The R User Conference 2013

July 10-12 2013

University of Castilla-La Mancha, Albacete, Spain

Supported by the R Foundation for Statistical Computing







- **Directions in Statistical Computing**

- Desarrolladores de software estadístico e investigadores en cómputo estadístico
- No centrado únicamente en R
- Serie histórica:

DSC 1999, Vienna, Austria

DSC 2001, Vienna, Austria

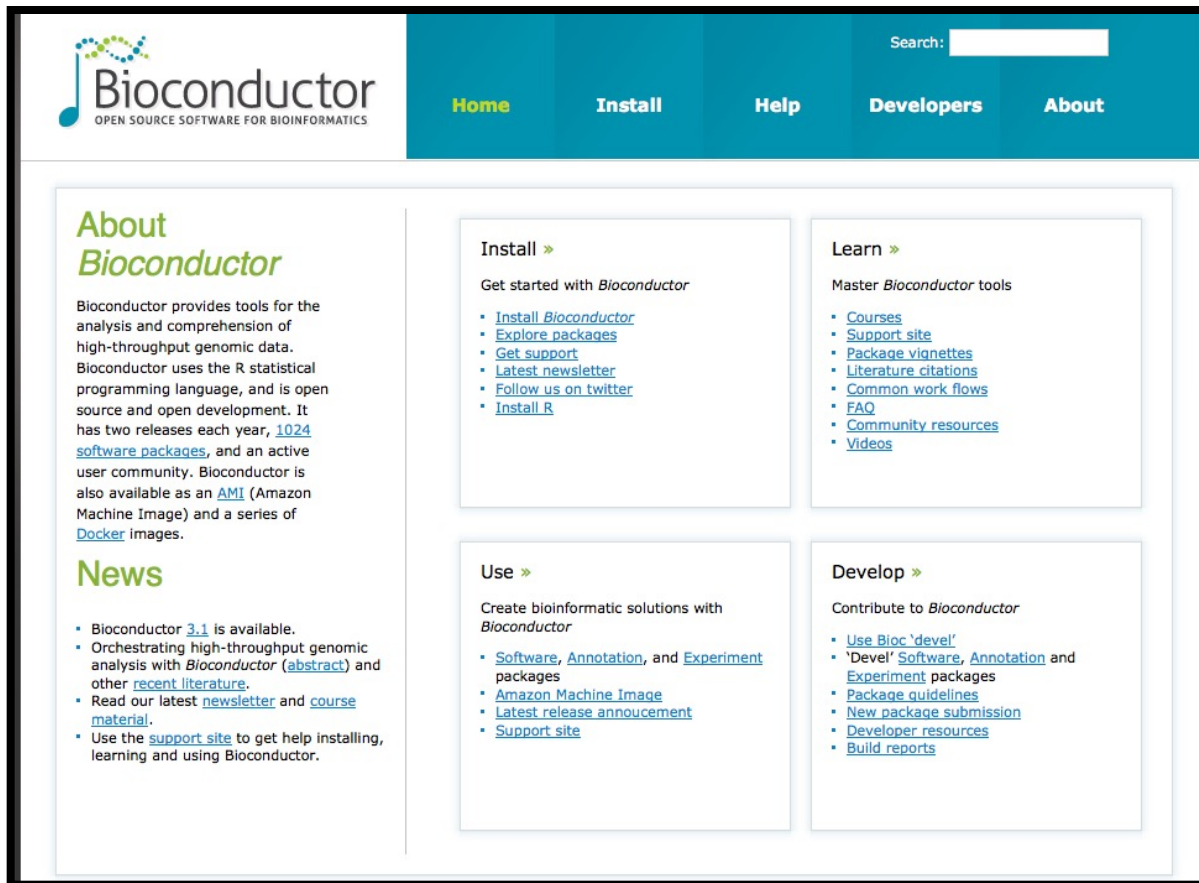
DSC 2003, Vienna, Austria

DSC 2005, Seattle, USA:

DSC 2007, Auckland, New Zealand

DSC 2009, Copenhagen, Denmark

Open source, open development software project to provide tools for the analysis and comprehension of high-throughput genomic data.



The screenshot shows the Bioconductor website interface. At the top, there is a navigation bar with links: Home, Install, Help, Developers, and About. A search bar is located on the right side of the navigation bar. The main content area is divided into several sections:

- About Bioconductor:** A section on the left providing an overview of the project, its goals, and its history. It mentions that Bioconductor provides tools for the analysis and comprehension of high-throughput genomic data and is open source and open development.
- Install:** A section in the top right providing links to get started with Bioconductor, including links to the installation guide, explore packages, get support, latest newsletter, follow us on twitter, and install R.
- Learn:** A section in the bottom right providing links to master Bioconductor tools, including courses, support site, package vignettes, literature citations, common work flows, FAQ, community resources, and videos.
- Use:** A section in the bottom left providing links to create bioinformatic solutions with Bioconductor, including software, annotation, and experiment packages, Amazon Machine Image, latest release announcement, and support site.
- Develop:** A section in the bottom right providing links to contribute to Bioconductor, including use Bioc 'devel', 'devel' software, annotation and experiment packages, package guidelines, new package submission, developer resources, and build reports.

<http://www.bioconductor.org/>

Package 'EBImage'

August 12, 2015

Version 4.10.1

Title Image processing and analysis toolbox for R

Encoding UTF-8

Author Andrzej Oleś, Gregoire Pau, Mike Smith, Oleg Sklyar, Wolfgang Huber, with contributions from Joseph Barry and Philip A. Marais

Maintainer Andrzej Oleś <andrzej.oles@embl.de>

Depends

Imports BiocGenerics (>= 0.7.1), methods, graphics, grDevices, stats, abind, tiff, jpeg, png, locfit, fftwtools (>= 0.9-7)

Suggests BiocStyle

Description EBImage provides general purpose functionality for image processing and analysis of microscopy-based cellular assays, EBImage of-
tract quantitative cellular descriptors. This allows the automa-
programming language and facili-
R environment for signal processing, statistical modeling, ma-
on.

pbdR: programming with big data in R

Simplifying Scalability

[About](#) [Packages](#) [Download](#) [Install](#) [Tutorials](#) [FAQ](#) [Misc](#)

Overview

The "Programming with Big Data in R" project (pbdR) enables high-level distributed data parallelism in R, so that it can easily utilize large HPC platforms with thousands of cores, making the R language scale to unparalleled heights. We interpret big data quite literally to mean that its size requires parallel processing either because it does not fit in the memory of a single multicore machine or because we need to make its processing time tolerable.

We achieve this, in part, by providing a simple interface to scalable, high performance libraries, such as MPI, ScaLAPACK, and NetCDF4. The routines in these libraries are engaged through R's classes and methods, so that the R language syntax is largely preserved, but with new, scalable, compiled code underneath. Most of the cumbersome distributed details are abstracted away for the user, although they are readily accessible should the user desire them.

We provide several R packages which focus on analyzing big data, especially on large distributed machines. The packages are meant to be used in the Single Program/Multiple Data (SPMD) programming model (see also), with special focus on large scale computing clusters. Packages and instructions for installation are available on the [downloads page](#).



RDataMining.com: R and Data Mining

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Introduction to Data Mining with R
R Reference Card for Data Mining
R and Data Mining: Examples and Case Studies
Introduction to Data Mining with R and Data Import/Export in R
Data Exploration and Visualization with R
Regression and Classification with R
Data Clustering with R
Association Rule Mining with R
Text Mining with R: Twitter Data Analysis
Time Series Analysis and Mining with R
Examples
Data Exploration
Decision Trees
Random Forest
k-means Clustering
Hierarchical Clustering
Outlier Detection
Time Series Forecasting
Time Series Analysis
Time Series Clustering and Classification
Association Rules
Text Mining
Twitter Follower Map
Social Network Analysis
Multidimensional Scaling (MDS)
Principal Component Analysis (PCA)
Parallel Computing
Other Examples
Big Data
Big Data Platforms

This website presents documents, examples, tutorials and resources on R and data mining.

Documents on Data Mining with R

- [R Reference Card for Data Mining](#)
- [R and Data Mining: Examples and Case Studies](#)
- [Introduction to Data Mining with R](#)
- [RDataMining slides series on](#)
 - [Introduction to Data Mining with R and Data Import/Export in R](#)
 - [Data Exploration and Visualization with R](#)
 - [Regression and Classification with R](#)
 - [Data Clustering with R](#)
 - [Association Rule Mining with R](#)
 - [Text Mining with R: Twitter Data Analysis](#), and
 - [Time Series Analysis and Mining with R](#)

Examples on Data Mining with R

- [data exploration](#)
- [decision trees](#)
- [k-means clustering and hierarchical clustering](#)
- [outlier detection](#)
- [time series decomposition and forecasting](#)
- [time series clustering and classification](#)
- [association rules](#)
- [text mining](#)
- [social network analysis](#)
- [Multidimensional Scaling \(MDS\)](#)
- [parallel computing](#)
- [many examples from other websites](#)

Search this site



News
[www2.rdatamining.com: a mirror site for Chinese users](#)
RDataMining.com now has a mirror website at <http://www2.rdatamining.com>. Users in China can download RDataMining documents, code and data at above mirror site, if no access to ...
Posted Aug 5, 2015, 5:31 AM by Yanchang Zhao
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AusDM 2015
Sydney, 8-9 August 2015

BIGIT
BIGDATA
9th - 10th November 2015
Abu Dhabi, UAE

BIGIT
BIGDATA
TECHNOLOGY SHOW
8-9 OCTOBER 2015

Survey analysis in R

This is the homepage for the "[survey](#)" package, which provides facilities in R for analyzing data from complex surveys. The current version is 3.29. A much earlier version (2.2) was published in [Journal of Statistical Software](#)

An experimental package for very large surveys such as the American Community Survey can be found [here](#)

A port of a much older version of the survey package (version 3.6-8) to S-PLUS 8.0 is available from [CSAN](#) (thanks to Patrick Abouyou at Insightful).

Features:

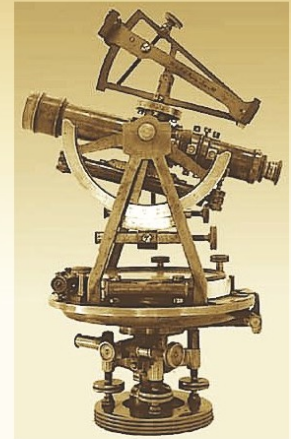
- Means, totals, ratios, quantiles, contingency tables, regression models, loglinear models, survival curves, rank tests, for the whole sample and for domains.
- Variances by Taylor linearization or by replicate weights (BRR, jackknife, bootstrap, multistage bootstrap, or user-supplied)
- Multistage sampling with or without replacement.
- PPS sampling with or without replacement: Horvitz-Thompson and Yates-Grundy estimators and a range of approximations.
- Post-stratification, generalized raking/calibration, GREG estimation, trimming of weights.
- Two-phase designs. Estimated weights for augmented IPW estimators.
- Graphics
- Support for using multiply imputed data
- Database-backed design objects for large data sets (now with replicate weights, too)
- Some support for parallel processing on multicore computers.
- Multivariate analysis: principal components, factor analysis (experimental).
- Likelihood ratio (Rao-Scott) tests for glms, Cox models, loglinear models.

The [NEWS](#) file gives a history of features and bug fixes.

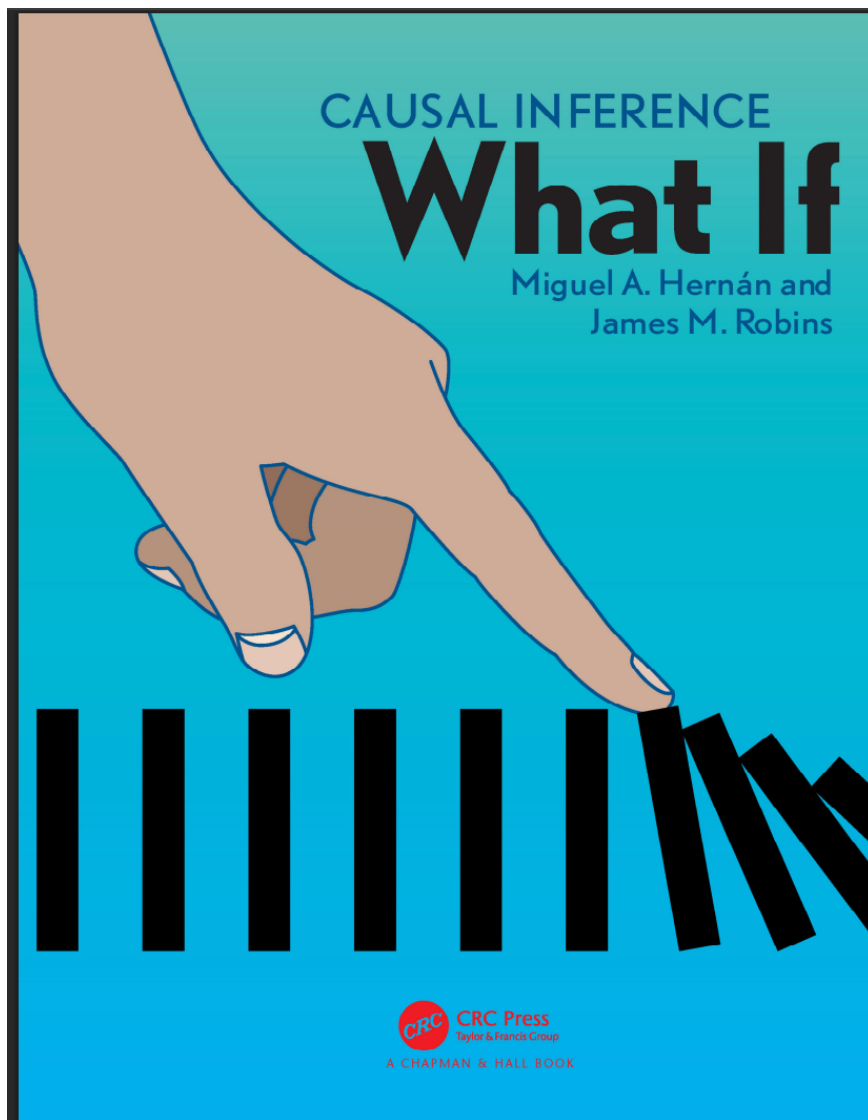
Comparison shopping:

Alan Zaslavsky keeps a comprehensive [list of survey analysis software](#) for the ASA Section on Survey Research Methods.

User-generated ratings and reviews of this package (and others) at [crantastic](#).

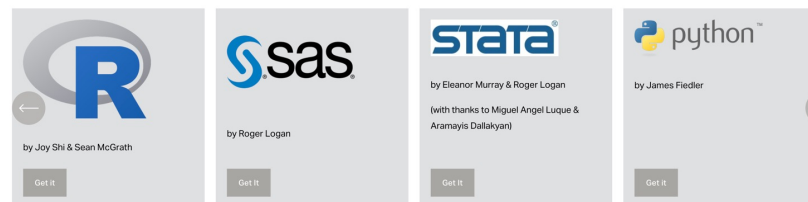


<http://r-survey.r-forge.r-project.org/survey/>



<https://miguelhernan.org/whatifbook>

The code used in the *What If* book



Methodology article

Open Access

Gene selection and classification of microarray data using random forest

Ramón Díaz-Uriarte^{*1} and Sara Álvarez de Andrés²

Address: ¹Bioinformatics Unit, Biotechnology Programme, Spanish National Cancer Centre (CNIO), Melchor Fernández Almagro 3, Madrid, 28029, Spain and ²Cytogenetics Unit, Biotechnology Programme, Spanish National Cancer Centre (CNIO), Melchor Fernández Almagro 3, Madrid, 28029, Spain

Email: Ramón Díaz-Uriarte^{*} - rdiaz@igarto.org; Sara Álvarez de Andrés - salv@igarto.org

^{*} Corresponding author

Software

SignS: a parallelized, open-source, freely available, web-based tool for gene selection and molecular signatures for survival analysis of censored data

Ramon Diaz-Uriarte

Address: Statistical Computing Team, Structural Biology and Biocomputing Programme, Spanish National Cancer Centre (CNIO), Melchor Fernández Almagro 3, Madrid, 28029, Spain

Email: Ramon Diaz-Uriarte - rdiaz02@gmail.com

Published: 21 January 2008

BMC Bioinformatics 2008, 9:30 doi:10.1186/1471-2105-9-30

This article is available from: <http://www.biomedcentral.com/1471-2105/9/30>

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Received
Accepted

Research article

The presence of the pilus locus is a clonal property among pneumococcal invasive isolates

Sandra I Aguiar, Isa Serrano, Francisco R Pinto, José Melo-Cristino and

Science of the Total Environment 435–436 (2012) 66–73



Contents lists available at SciVerse ScienceDirect

Science of the Total Environment

journal homepage: www.elsevier.com/locate/scitotenv



Proximity to mining industry and cancer mortality

Pablo Fernández-Navarro^{a,b,*}, Javier García-Pérez^{a,b}, Rebeca Ramis^{a,b,c}, Elena Boldo^{a,b}, Gonzalo López-Abente^{a,b}

^a Cancer and Environmental Epidemiology Unit, National Center for Epidemiology, Carlos III Institute of Health, Avda. Monforte de Lemos, 5, 28029 Madrid, Spain

^b CIBER Epidemiología y Salud Pública (CIBERESP), Spain

^c School of Health and Medicine, Lancaster University, UK

Environment International 40 (2012) 116–127



Contents lists available at ScienceDirect

Environment International

journal homepage: www.elsevier.com/locate/envint



Risk of dying of cancer in the vicinity of multiple pollutant sources associated with the metal industry

Javier García-Pérez^{a,b,*}, María Felicitas López-Cima^{a,b,c}, Marina Pollán^{a,b}, Beatriz Pérez-Gómez^{a,b}, Nuria Aragonés^{a,b}, Pablo Fernández-Navarro^{a,b}, Rebeca Ramis^{a,b}, Gonzalo López-Abente^{a,b}

^a Cancer and Environmental Epidemiology Unit, National Center for Epidemiology, Carlos III Institute of Health, Avda. Monforte de Lemos, 5, 28029 Madrid, Spain

^b CIBER Epidemiología y Salud Pública (CIBERESP), Spain

^c Molecular Epidemiology of Cancer Unit, University Institute of Oncology, University of Oviedo, C/Fernando Bongera, s/n, 33006 Oviedo, Spain

Molecular, Faculdade de Medicina Universidade de Lisboa, Lisboa, Portugal
aserrano@fm.ul.pt; Francisco R Pinto - fpinto@fm.ul.pt; José Melo-Cristino - jmelo@fm.ul.pt

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471-2180/8/41

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9. Obtención e instalación

- Depende del sistema operativo, pero todo se puede encontrar en <http://cran.es.r-project.org/>
- **Windows:** se puede obtener un ejecutable (al ejecutar el archivo se instalará el sistema base y los paquetes recomendados).
- **GNU/Linux:** (dos opciones)
 - a. Obtener el R-x.y.z.tar.gz y compilar desde las fuentes. También bajar los paquetes adicionales y instalarlos.
 - b. Obtener binarios (ej., *.deb para Debian, *.rpm para RedHat, SuSE, Mandrake).".
- **Mac:** R-3.2.1.pkg (requires [XQuartz](#) to be installed)



The R Project for Statistical Computing

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Getting Started

R is a free software environment for statistical computing and graphics. It compiles and runs on a wide variety of UNIX platforms, Windows and MacOS. To [download R](#), please choose your preferred CRAN mirror.

If you have questions about R like how to download and install the software, or what the license terms are, please read our [answers to frequently asked questions](#) before you send an email.

News

- R version 3.2.2 (Fire Safety) prerelease versions** will appear starting 2015-08-04. Final release is scheduled for 2015-08-14.
- The R Journal Volume 7/1** is available.
- R version 3.2.1 (World-Famous Astronaut)** has been released on 2015-06-18.
- R version 3.1.3 (Smooth Sidewalk)** has been released on 2015-03-09.
- useR! 2015**, will take place at the University of Aalborg, Denmark, June 30 - July 3, 2015.
- useR! 2014**, took place at the University of California, Los Angeles, USA June 2-4, 2014.

CRAN Mirrors

The Comprehensive R Archive Network is available at the following URLs, please choose a location close to you. Some statistics on the status of the mirrors are available [here](#).

0-Cloud	https://cran.rstudio.com/	Rstudio, automatic redirection to servers worldwide
Algeria	http://cran.rstudio.com/	Rstudio, automatic redirection to servers worldwide
Argentina	http://mirror.fcaglp.unlp.edu.ar/CRAN/	University of Science and Technology
Australia	http://cran.csiro.au/	CSIRO
Austria	http://cran.ms.unimcb.edu.au/	University of Vienna
Austria	https://cran.r-project.org/	
Belgium	http://cran.at.r-project.org/	
Brazil	http://www.freestats.org/cran/	
Brazil	http://ibgph.usesc.br/mirrors/cran/	Universidade Estadual de Santa Cruz
Brazil	http://cran-fc3h.ufpe.br/	
Brazil	http://cran.fiocruz.br/	
Brazil	http://www.vps.fmvz.usp.br/CRAN/	
Brazil	http://bricor.esalq.usp.br/CRAN/	
Canada	http://cran.stat.sfu.ca/	Simon Fraser University, Burnaby
Canada	http://mirror.its.dal.ca/cran/	Dalhousie University, Halifax
Canada	http://cran.usnsl.usc.edu/	University of Toronto
Canada	http://cran.skaakafory.com/	iWeb, Montreal
Canada	http://cran.parentingamerica.com/	iWeb, Montreal
Chile	http://dichetnet.mat.puc.cl/	Pontificia Universidad Catolica de Chile, Santiago
China	http://mirror.bjtu.edu.cn/cran/	Beijing Jiaotong University, Beijing
China	http://mirrors.opencas.cn/cran/	Chinese Academy of Sciences, Beijing
China	http://mirrors.tuna.tsinghua.edu.cn/CRAN/	TUNA Team, Tsinghua University
China	http://mirrors.tuna.tsinghua.edu.cn/CRAN/	TUNA Team, Tsinghua University
China	http://mirrors.usc.edu.cn/CRAN/	University of Science and Technology of China
China	http://mirrors.usc.edu.cn/CRAN/	University of Science and Technology of China
China	http://mirrors.xmu.edu.cn/CRAN/	Xiamen University
Colombia	http://www.icesi.edu.co/CRAN/	Icesi University
Colombia	http://www.icesi.edu.co/CRAN/	Icesi University



The Comprehensive R Archive Network

Download and Install R

Precompiled binary distributions of the base system and contributed packages, **Windows** and **Mac** users most likely want one of these versions of R:

- [Download R for Linux](#)
- [Download R for \(Mac\) OS X](#)
- [Download R for Windows](#)

R is part of many Linux distributions, you should check with your Linux package management system in addition to the link above.

Source Code for all Platforms

Windows and Mac users most likely want to download the precompiled binaries listed in the upper box, not the source code. The sources have to be compiled before you can use them. If you do not know what this means, you probably do not want to do it!

- The latest release (2015-06-18, World-Famous Astronaut) [R-3.2.1.tar.gz](#), read [what's new](#) in the latest version.
- Sources of [R alpha](#) and [beta releases](#) (daily snapshots, created only in time periods before a planned release).
- Daily snapshots of current patched and development versions are [available here](#). Please read about [new features and bug fixes](#) before filing corresponding feature requests or bug reports.
- Source code of older versions of R is [available here](#).
- Contributed extension [packages](#)

Questions About R

- If you have questions about R like how to download and install the software, or what the license terms are, please read our [answers to frequently asked questions](#) before you send an email.

What are R and CRAN?

R is "GNU S", a freely available language and environment for statistical computing and graphics which provides a wide variety of statistical and graphical techniques: linear and nonlinear modelling, statistical tests, time series analysis, classification, clustering, etc. Please consult the [R project homepage](#) for further information.

CRAN is a network of ftp and web servers around the world that store identical, up-to-date, versions of code and documentation for R. Please use the CRAN [mirror](#) nearest to you to minimize network load.

Submitting to CRAN

To "submit" a package to CRAN, check that your submission meets the [CRAN Repository Policy](#) and then use the [web form](#).

If this fails, upload to <ftp://CRAN.R-project.org/incoming/> and send an email to CRAN@R-project.org following the policy. Please do not attach submissions to emails, because this will clutter up the mailboxes of half a dozen people.

Note that we generally do not accept submissions of precompiled binaries due to security reasons. All binary distribution listed above are compiled by selected maintainers, who are in charge for all binaries of their platform, respectively.

This server is hosted by the [Institute for Statistics and Mathematics of WU \(Wirtschaftsuniversität Wien\)](#).

LO INSTALAMOS EN EL ORDENADOR

Consideraciones

- Coexisten versiones de R (cuidado en Linux y Mac)
- Los paquetes instalados se almacenan en las carpetas de la versión de R donde hayamos realizado la instalación

10. Documentación (1/4)

A) Manuales incluidos en la instalación

A) Documentación general

A) Estadística

10. Documentación (2/4)

- A) Los **manuales de R incluidos** en todas las instalaciones son (*C:\Program Files\R\R-2.6.2\doc\html\index.html*):
 - An introduction to R.*
 - Writing R extensions.*
 - R data import/export.*
 - The R language definition.*
 - R installation and administration.*
- B) **Documentación general (CRAN):**
 - R para principiantes, de E. Paradis.*
http://cran.at.r-project.org/doc/contrib/rdebuts_es.pdf
 - A guide for the unwilling S user, de P. Burns.*
http://cran.at.r-project.org/doc/contrib/Burns-unwilling_S.pdf
 - R Graphics, de Paul Murrell. S Programming, de W. Venables y B. Ripley.*
<http://www.stat.auckland.ac.nz/~paul/RGraphics/rgraphics.html>

10. Documentación (3/4)

- **C) Estadística (CRAN y otros):**

- simpleR *Using R for Introductory Statistics* de John Verzani.

- <http://cran.at.r-project.org/doc/contrib/Verzani-SimpleR.pdf>

- Introductory statistics with R* de P. Dalgaard.

- <http://staff.pubhealth.ku.dk/~pd/ISwR.html>

- An R and S-PLUS companion to applied regression* de J. Fox.

- <http://socserv.mcmaster.ca/jfox/Books/Companion/index.html>

- Modern applied statistics with S, 4th ed.* de W. Venables y B. Ripley.

- <http://www.stats.ox.ac.uk/pub/MASS4/>

- Practical regression and ANOVA using R* de J. Faraway.

- <http://cran.r-project.org/doc/contrib/Faraway-PRA.pdf>

10. Documentación (4/4)

-Otros documentos:

<http://www.r-project.org/other-docs.html>

-*Mixed-effects models in S and S-PLUS* de J. Pinheiro y D. Bates.

-*Regression modelling strategies* de F. Harrell.

-*Modelos Lineales* de F. Carmona.

-Documentación sobre análisis para datos categóricos.

<http://www.stat.ufl.edu/~aa/cda/cda.html>

-*Modeling survival data: extending the Cox model* de T. M. Therneau y P. M. Grambsch.

-Documentos varios en la página de J. Fox

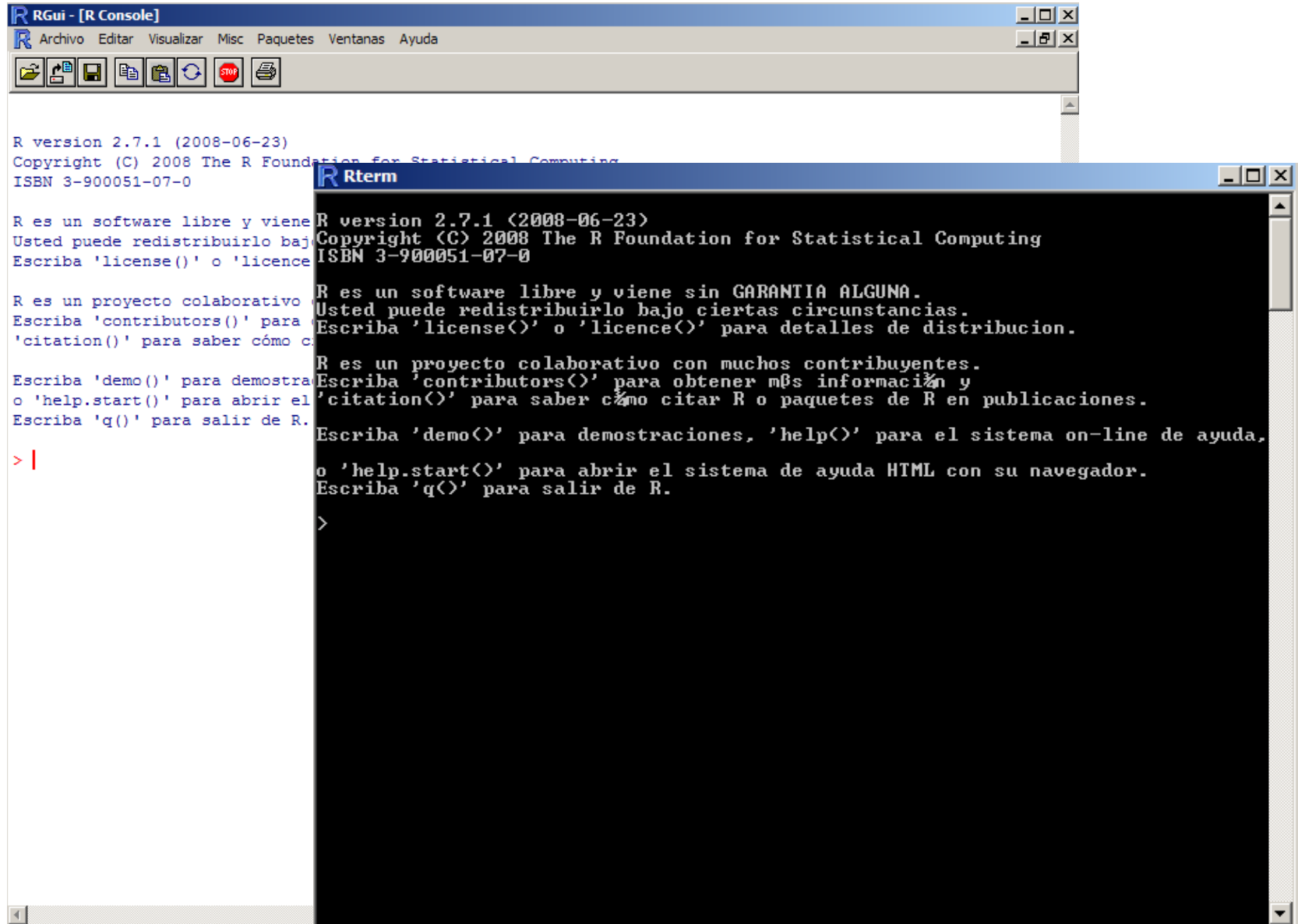
<http://cran.r-project.org/doc/contrib/Fox-Companion/appendix.html>

11. Inicio de sesión con R



- **Windows:**
 - a. Hacer doble-click en el icono. Se abrirá “Rgui”.
 - b. Desde una “ventana del sistema” ejecutar Rterm
 - c. Desde Inicio-Ejecutar (cmd...R)
 - d. “C:\Archivos de programa\R\R-2.7.1\bin\Rterm.exe”
 - e. Iniciar R desde Tinn-R, XEmacs u otro editor.

- **GNU/Linux:**
 - a. Escribir “R” en una shell.
 - b. Iniciar R desde (X)Emacs (M-X R) u otro editor.
 - c. Otros



The image shows two overlapping windows from the R statistical software environment. The background window is 'RGui - [R Console]', which has a menu bar (Archivo, Editar, Visualizar, Misc, Paquetes, Ventanas, Ayuda) and a toolbar with icons for file operations and execution. The foreground window is 'Rterm', a terminal window with a black background and white text. Both windows display the R version 2.7.1 startup message and a list of help topics. The Rterm window also includes the phrase 'sin GARANTIA ALGUNA'.

```
R version 2.7.1 (2008-06-23)
Copyright (C) 2008 The R Foundation for Statistical Computing
ISBN 3-900051-07-0

R es un software libre y viene sin GARANTIA ALGUNA.
Usted puede redistribuirlo bajo ciertas circunstancias.
Escriba 'license()' o 'licence()' para detalles de distribucion.

R es un proyecto colaborativo con muchos contribuyentes.
Escriba 'contributors()' para obtener m s informaci n y
'citation()' para saber c mo citar R o paquetes de R en publicaciones.

Escriba 'demo()' para demostraciones, 'help()' para el sistema on-line de ayuda,
o 'help.start()' para abrir el sistema de ayuda HTML con su navegador.
Escriba 'q()' para salir de R.

> |
```

```
R version 2.7.1 (2008-06-23)
Copyright (C) 2008 The R Foundation for Statistical Computing
ISBN 3-900051-07-0

R es un software libre y viene sin GARANTIA ALGUNA.
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o 'help.start()' para abrir el sistema de ayuda HTML con su navegador.
Escriba 'q()' para salir de R.

>
```

Guardar área
de trabajo

Copiar

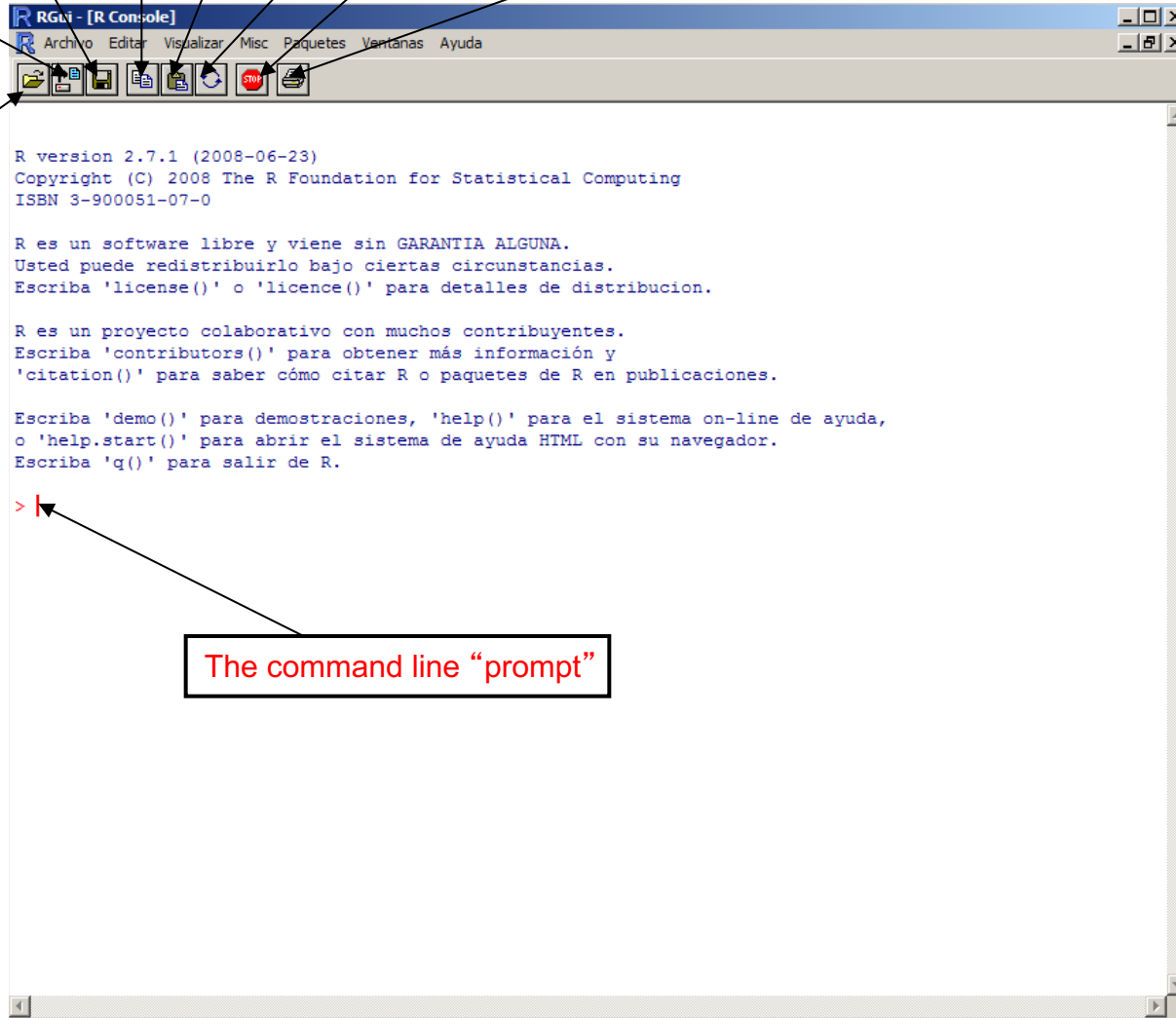
Pegar

Copiar y
pegar

Parar
computación
actual

Imprimir

Cargar área
de trabajo



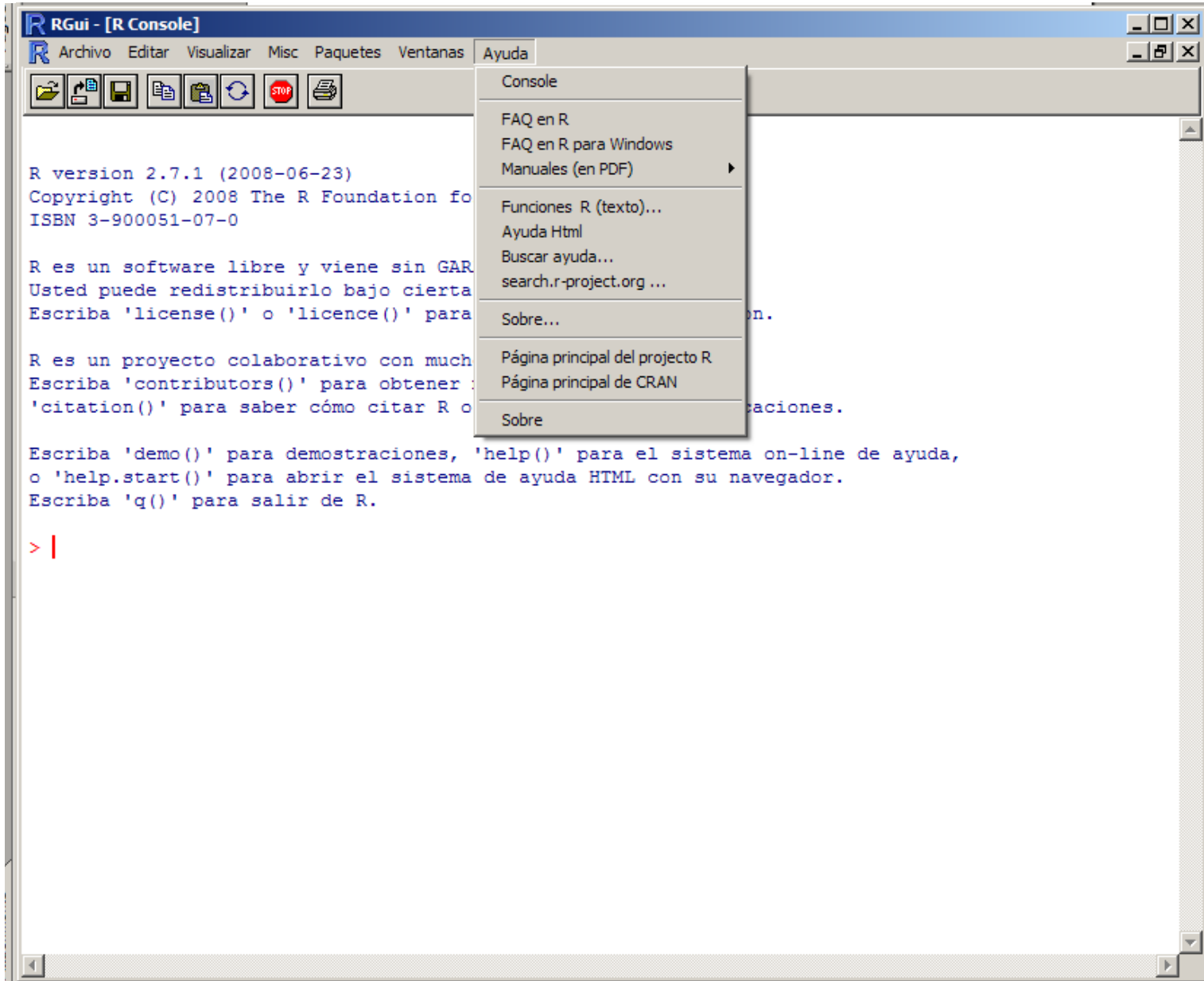
The command line “prompt”

Abrir script

WORKSPACE O ÁREA DE TRABAJO



- Espacio de memoria donde se encuentran almacenados los objetos CREADOS o CARGADOS para una sesión
- Espacio de memoria físicamente localizado en Memoria RAM
- Se elimina al cerrar sesión de R





```
R RGui - [R Console]
R Archivo Editar Visualizar Misc Paquetes Ventanas Ayuda

R version 2.7.1 (2008-06-23)
Copyright (C) 2008 The R Foundation for Statistical Computing
ISBN 3-900051-07-0

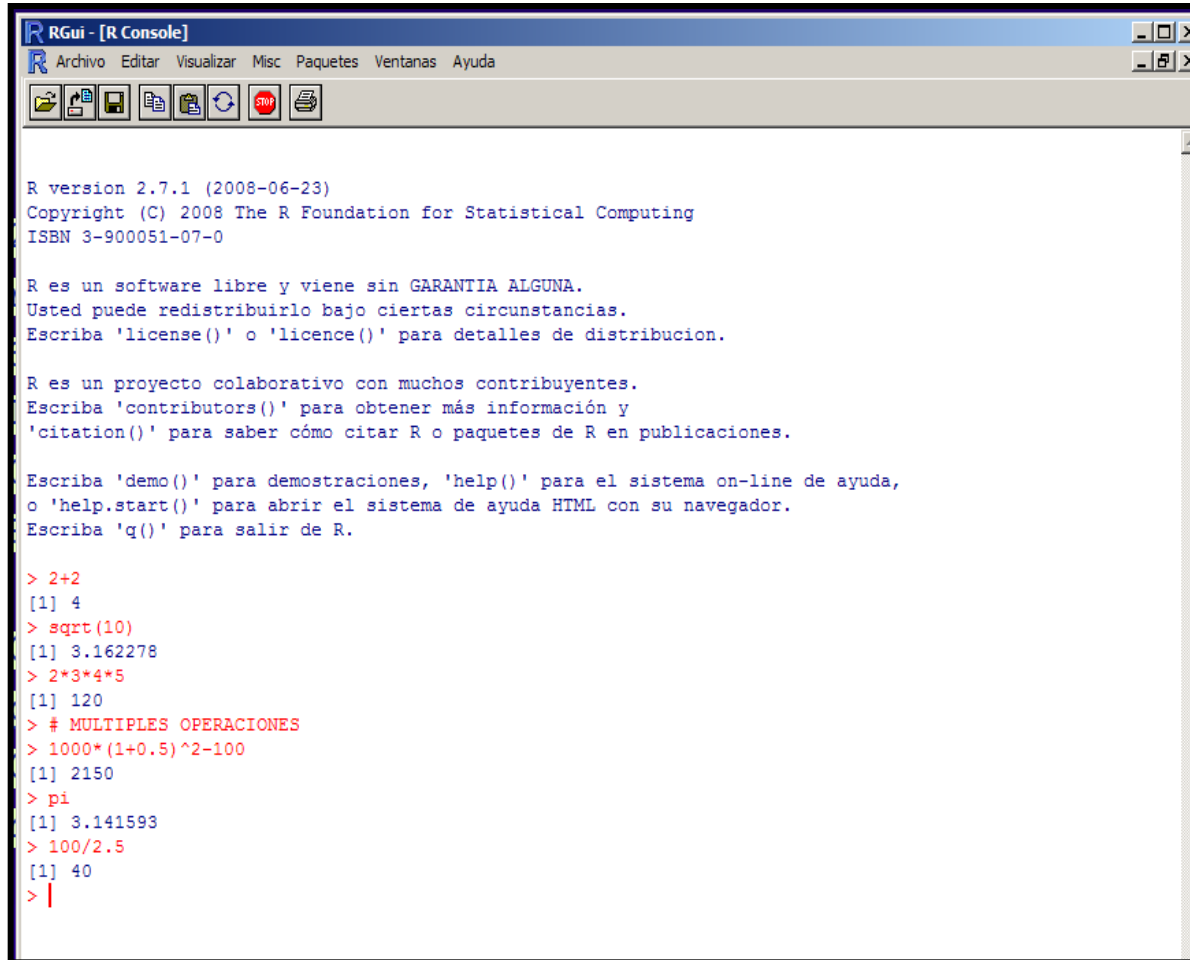
R es un software libre y viene sin GARANTIA ALGUNA.
Usted puede redistribuirlo bajo ciertas circunstancias.
Escriba 'license()' o 'licence()' para detalles de distribucion.

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Escriba 'demo()' para demostraciones, 'help()' para el sistema on-line de ayuda,
o 'help.start()' para abrir el sistema de ayuda HTML con su navegador.
Escriba 'q()' para salir de R.

> rnorm(5) # 5 números aleatorios de una normal (mean= 0, sd = 1)
[1] 0.09385335 -0.75646882 -1.65607431 -1.51165259 -0.13422251
> x <- rnorm(5) # asignamos unos números a un objeto (un vector) x
> x
[1] 1.22792041 0.58418161 -0.50232603 -0.85626820 -0.05495992
> summary(x) ## muestra un resumen de x (un summary "inteligente")
      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
-0.85630 -0.50230 -0.05496  0.07971  0.58420  1.22800
> w <- summary(x)
> w
      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
-0.85630 -0.50230 -0.05496  0.07971  0.58420  1.22800
> print(w)
      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
-0.85630 -0.50230 -0.05496  0.07971  0.58420  1.22800
> |
```

R como calculadora



```
RGui - [R Console]
Archivo  Editar  Visualizar  Misc  Paquetes  Ventanas  Ayuda

R version 2.7.1 (2008-06-23)
Copyright (C) 2008 The R Foundation for Statistical Computing
ISBN 3-900051-07-0

R es un software libre y viene sin GARANTIA ALGUNA.
Usted puede redistribuirlo bajo ciertas circunstancias.
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'citation()' para saber cómo citar R o paquetes de R en publicaciones.

Escriba 'demo()' para demostraciones, 'help()' para el sistema on-line de ayuda,
o 'help.start()' para abrir el sistema de ayuda HTML con su navegador.
Escriba 'q()' para salir de R.

> 2+2
[1] 4
> sqrt(10)
[1] 3.162278
> 2*3*4*5
[1] 120
> # MULTIPLES OPERACIONES
> 1000*(1+0.5)^2-100
[1] 2150
> pi
[1] 3.141593
> 100/2.5
[1] 40
> |
```

PRACTICAMOS

12. Ayudas

- Ayuda fuera del programa
- Ayuda dentro del programa

Ayuda fuera del programa

1. Frequently Asked Questions (**FAQ**).

<http://cran.r-project.org/faqs.html>

2. **Rtips** de Paul Johnson (“list of things I'm using to remember what I read in the email list”)

<http://pj.freefaculty.org/R/Rtips.html>

3. **R Help Center** (Institut de statistique, UCL (Belgium))

<http://www.stat.ucl.ac.be/ISdidactique/Rhelp/>

4. Las **e-mail lists**

<http://www.r-project.org/search.html>

<http://finzi.psych.upenn.edu/search.html>

Permiten hacer las búsquedas no sólo sobre las listas de e-mail sino también sobre la documentación (incluyendo paquetes).

Ayuda dentro del programa

Con comandos

>?rnorm

>?demo

>help.start()

>demo(graphics); demo(persp);

>?help.search

>demo(lm.glm)

>help.search("normal")

!Cuidado;

>?apropos

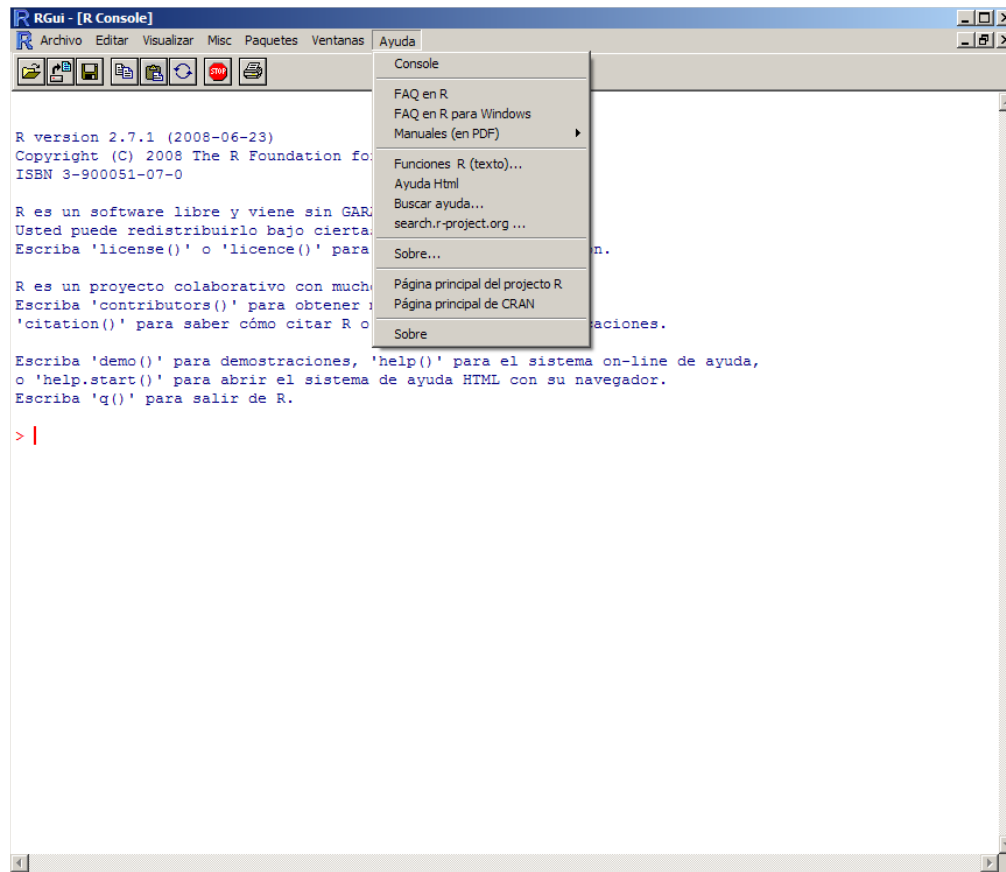
>?if # mal

>help("if")

>apropos("normal")

Ayuda dentro del programa

Desde el menu del programa





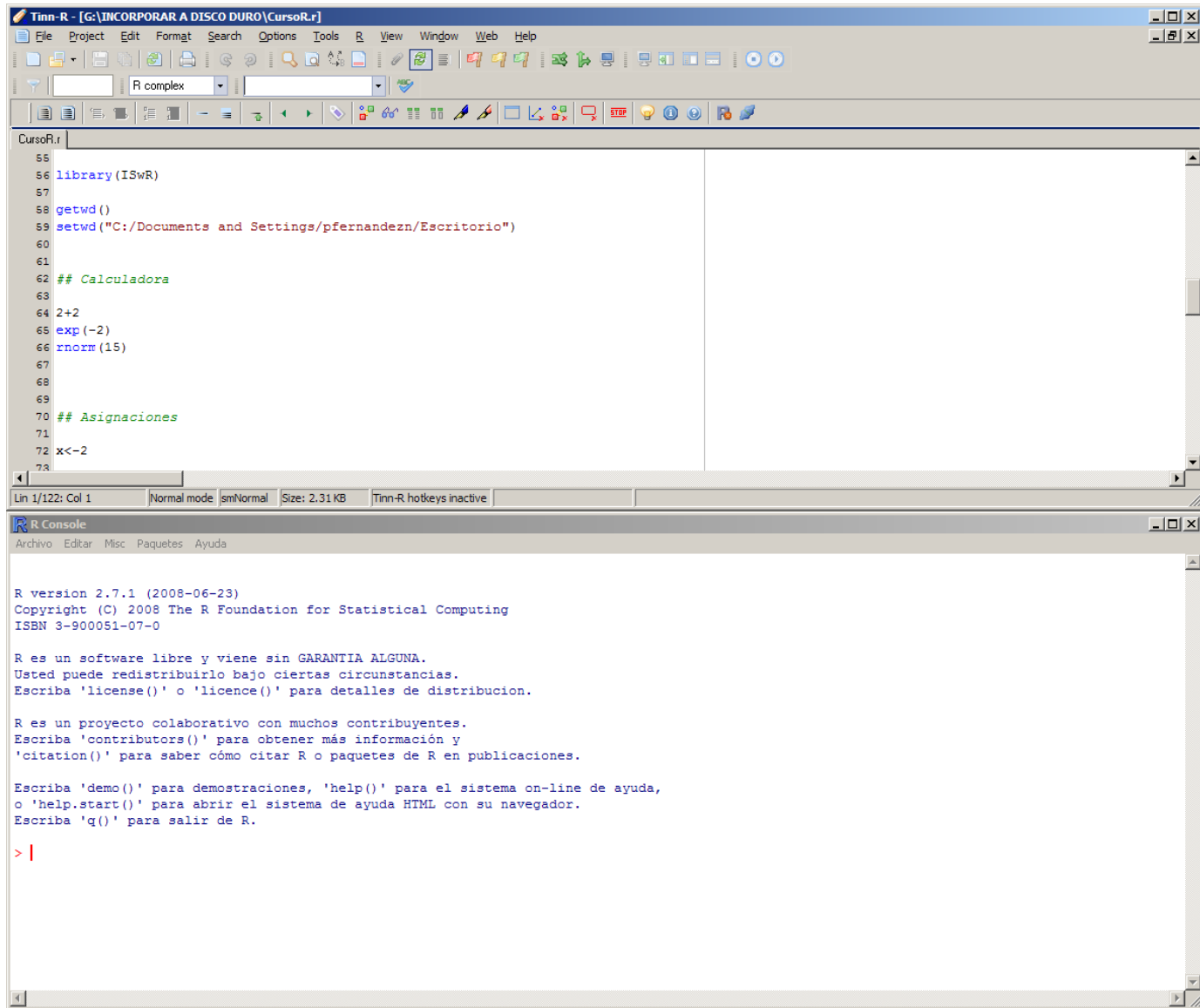
**PRACTICAMOS LAS AYUDAS FUERA Y
DENTRO DEL PROGRAMA**



13. Usar R desde un editor (1/2)

- **¿Por qué usar R desde un editor como texmate, Tinn-R, R-WinEdt o XEmacs?**
 - El uso de scripts y el mantenimiento del código ordenado y comentado es una “buena práctica estadística”.
 - Colores de sintaxis, completa paréntesis, etc.
 - Una interface común para otros paquetes estadísticos (ej., SAS, XLispStat, Arc, etc.), numéricos (ej., Octave) o procesador de texto (LaTeX).

TinnR



The screenshot shows the TinnR application window. The title bar reads "Tinn-R - [G:\INCORPORAR A DISCO DURO\CursoR.r]". The menu bar includes File, Project, Edit, Format, Search, Options, Tools, R, View, Window, Web, and Help. The toolbar contains various icons for file operations, editing, and running code. The main editor area displays the following R code:

```
55  
56 library(ISwR)  
57  
58 getwd()  
59 setwd("C:/Documents and Settings/pfernandezn/Escritorio")  
60  
61  
62 ## Calculadora  
63  
64 2+2  
65 exp(-2)  
66 rnorm(15)  
67  
68  
69  
70 ## Asignaciones  
71  
72 x<-2  
73
```

Below the editor is the R Console window. Its title bar reads "R Console". The menu bar includes Archivo, Editar, Misc, Paquetes, and Ayuda. The console displays the following text:

```
R version 2.7.1 (2008-06-23)  
Copyright (C) 2008 The R Foundation for Statistical Computing  
ISBN 3-900051-07-0  
  
R es un software libre y viene sin GARANTIA ALGUNA.  
Usted puede redistribuirlo bajo ciertas circunstancias.  
Escriba 'license()' o 'licence()' para detalles de distribucion.  
  
R es un proyecto colaborativo con muchos contribuyentes.  
Escriba 'contributors()' para obtener más información y  
'citation()' para saber cómo citar R o paquetes de R en publicaciones.  
  
Escriba 'demo()' para demostraciones, 'help()' para el sistema on-line de ayuda,  
o 'help.start()' para abrir el sistema de ayuda HTML con su navegador.  
Escriba 'q()' para salir de R.  
  
> |
```



textmate

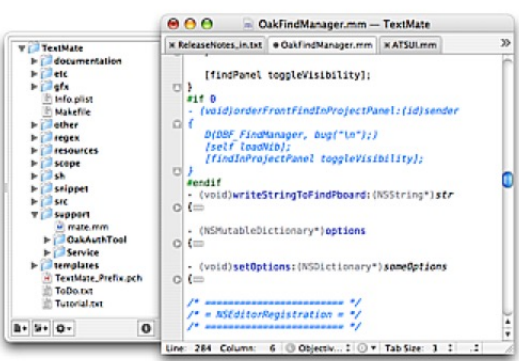
the missing editor

code and markup brought to the 21st century

[Intro](#)
[Blog](#)
[Manual](#)
[Wiki](#)
[Support](#)

TextMate brings Apple's approach to operating systems into the world of text editors. By bridging UNIX underpinnings and GUI, TextMate cherry-picks the best of both worlds to the benefit of expert scripters and novice users alike.

Whether you are a programmer or a designer, the production of code and markup is hard work. Without an editor dedicated to the task, it is also often cumbersome, overwhelming, and repetitive. Especially when you are dealing with a lot of files at once — like most projects do. TextMate puts you back in control, reduces the mental overhead, and turns manual work into something the computer does.




Created by a closet UNIX geek who was lured to the Mac platform by its ease of use and elegance, TextMate has been referred to as the culmination of Emacs and OS X and has resulted in countless requests for both a Windows and Linux port, but TextMate remains



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TextMate 2.0 beta
or 1.5 for OS X 10.4-6



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Documentation

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Watch screencasts featuring
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Read the TextMate book



Entorno de desarrollo integrado (IDE) para R que proporciona una interfaz gráfica para escribir, ejecutar y gestionar código, datos y resultados en un mismo lugar.



- **Editor de scripts** para escribir y ejecutar código R.
- **Consola integrada** para ejecutar comandos directamente.
- **Gestión del entorno de trabajo** (objetos, historial, variables).
- **Paneles para gráficos, archivos y paquetes.**
- **Herramientas para informes reproducibles** (R Markdown / Quarto).

<https://posit.co/download/rstudio-desktop/>